QUEEN'S UNIVERSITY BELFAST | QUEEN'S BUSINESS SCHOOL

Contents

Introduction

International retail expansion requires firms to decide not only whether to enter new markets, but also which markets offer the strongest balance of opportunity and risk. This makes market selection a strategic decision, as firms must compare countries across a range of economic, demographic, trade, and development conditions rather than relying on one indicator alone. Recent international market selection research shows that attractive markets should be assessed using multiple criteria, while location attractiveness literature suggests that market potential is increasingly shaped by broader structural conditions such as development quality, resilience, and connectivity (Andy et al., 2024, Francioni and Martín Martín, 2024)

This report applies a country-level quantitative analytics approach to assess market attractiveness for international retail expansion. Using a dataset containing indicators such as GDP, GDP per capita, imports, exports, unemployment, life expectancy, infant mortality, population growth, urban population growth, and business confidence, the analysis is divided into two stages. First, exploratory data analysis (EDA) is used to identify broad regional patterns and country-level differences that may matter for business users. Second, predictive data analysis (PDA) is used to test whether country performance can be estimated from the available indicators and to identify which variables contribute most strongly to those outcomes (Intern-report, 2026).

The report is designed to provide evidence-based insight for managerial decision-making. Rather than producing a single ranking in isolation, it uses visual analysis and predictive modelling to identify which countries appear most attractive, which factors explain that attractiveness, and which indicators are most useful for supporting future market screening decisions. This is consistent with the assignment aim of delivering country-level insight and practical recommendations for business users.

Part 1a: Exploratory Data Analysis (EDA)

The exploratory data analysis (EDA) was done to identify which markets appear most attractive for international retail expansion and to explain why. This stage follows a multi-criteria international retail expansion and to explain why. This stage follows a multi-criteria international market selection logic, where countries are assessed with a combination of demographic, trade, labour-market, economic and development indicators rather than a single metric. This approach is supported by recent international market selection research, which argues that foreign market decisions are strategic, complex, and shaped by multiple factors, distance, institutional conditions, and networks, rather than one universal selection rule.

The EDA was conducted in two stages. First, regional analysis was used to identify broad geographic patterns. Second, country-level analysis was used to examine variation within the stronger regions. This sequencing is justified by both the market selection and retail internationalisation literature. International market selection should not stop at broad market screening because opportunities and outcomes differ by unit of analysis, even within emerging market groups, attractiveness varies substantially across countries and influences retail internationalisation differently (Francioni and Martín Martín, 2024).

At the regional level, Europe and Asia emerged as the strongest regions overall, Europe performed more strongly on development-related indicators, including GDP per capita (see Fig 1), life expectancy, and infant mortality, suggesting a more mature, stable, lower-risk operating environment. Asia performed better on trade, net exports, demographic and urban growth, indicating stronger scale, market drive, and loner-term demand potential (see Appendix A, Figures A1, A2, A4, A10, A11, A12 and A15).

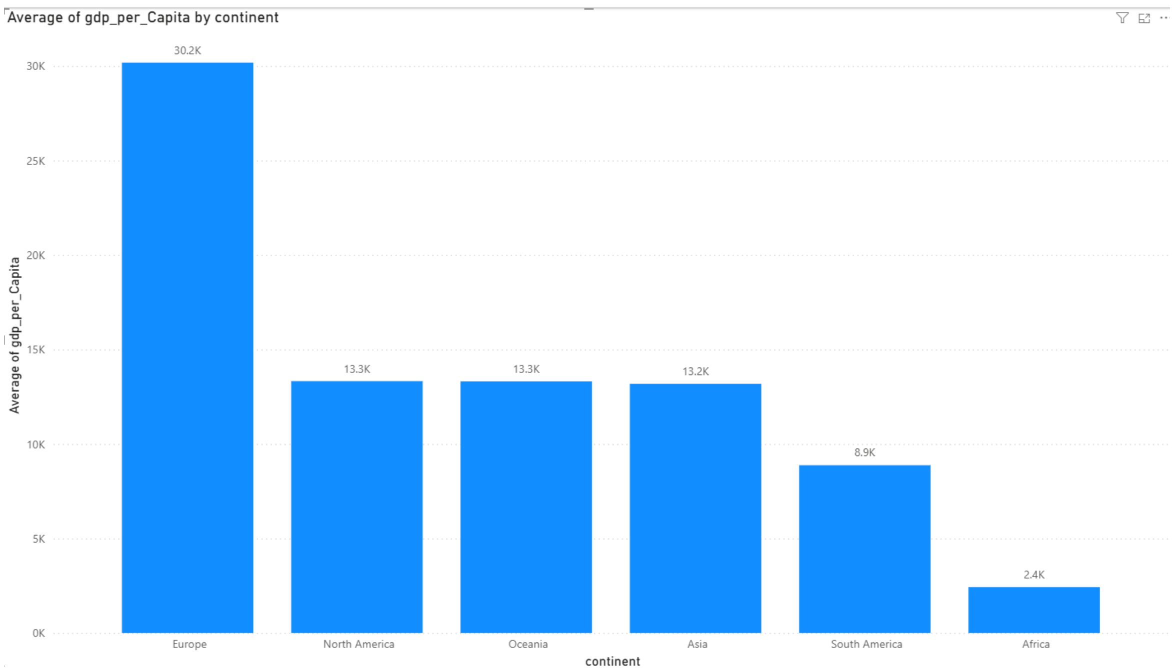

Figure 1: Average GDP per capita by continent, showing that Europe has the strongest overall purchasing power, while Asia performs at a lower average level but remains commercially important due to its scale.

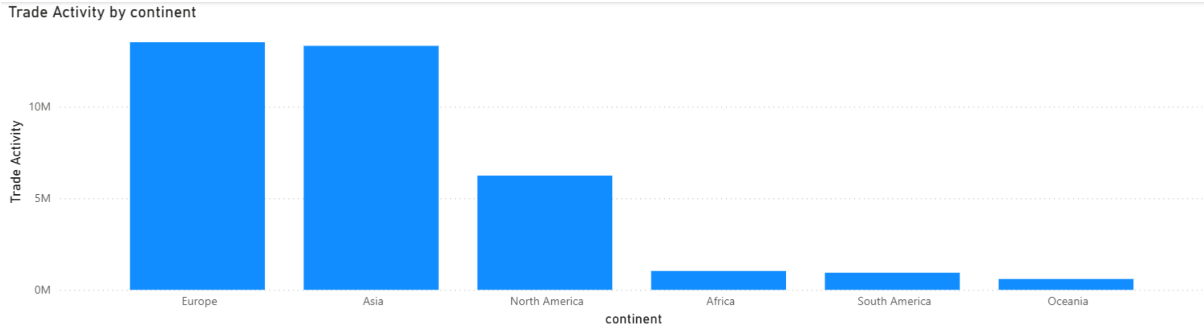

Figure 2: Trade activity by continent, highlighting that Asia and Europe are the most commercially connected regions and therefore the strongest candidates for further market investigation.

These broad patterns are consistent with the OECD's multidimensional view of attractiveness, which argues that location attractiveness should be understood though a combination of economic attraction, connectedness, well-being, and development conditions rather than a single indicator alone (OECD, 2024).

The country-level visuals showed that no single variable was sufficient to judge market attractiveness. Countries such as China, Germany, and Japan stood out for scale, particularly in GDP and trade measures, and therefore represent large, globally integrated markets (see Appendix B, Figures B8, B9 and B10). However, scale alone did not imply the best overall operating conditions. Other countries such as Iceland, Singapore, and Finland, appeared stronger on development and labour-related indicators, suggesting lower operating risk and stronger institutional conditions (see Appendix B, Figures B5 and B6). A third group, including Bahrain, Oman and Qatar, showed stronger lagged growth in population growth, indicating potential long-term market expansion (see Appendix B, Figure B7). This pattern supports the argument from recent location attractiveness research that country attractiveness is shaped not only by size and costs, but also by broader conditions such as resilience, human development, and long-run capability.

The EDA has showed that several trade-offs between variables. High GDP did not always coincide with lower unemployment, and stronger population did not imply better development outcomes. Similarly, trade surpluses and deficits could not be interpreted in isolation, because trade position alone did not map neatly onto overall market attractiveness (see Appendix B, Figures B1, to B4).

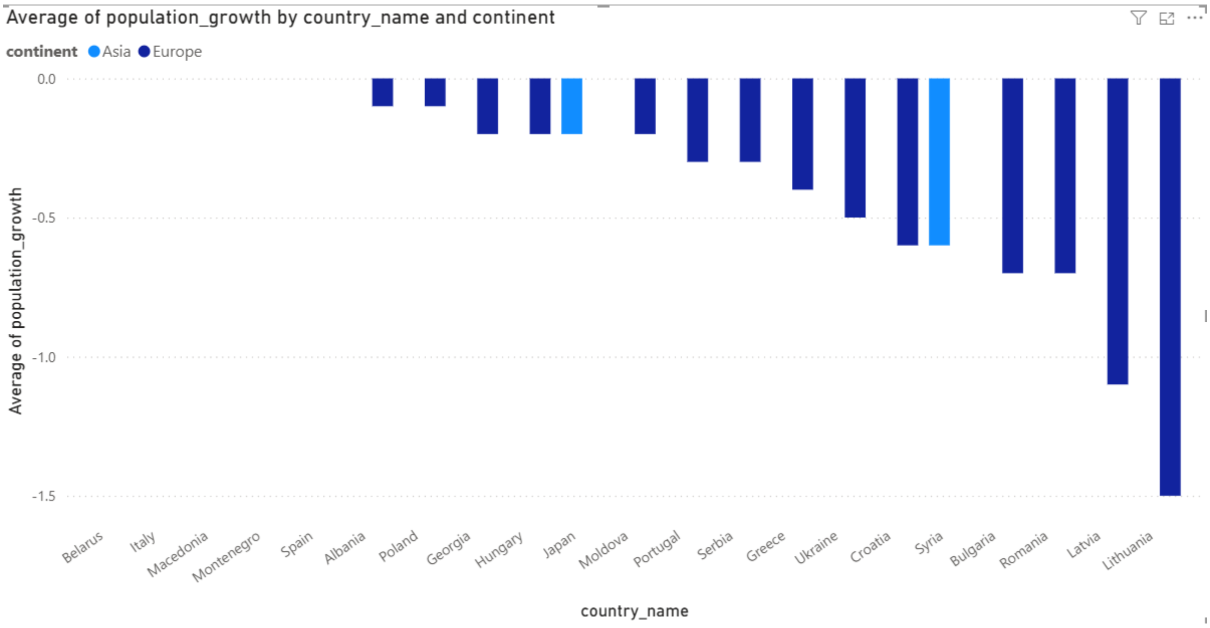

Figure 3: Country-level comparison of leading markets in Europe and Asia, showing that a small number of countries account for much of the region's economic and trade strength.

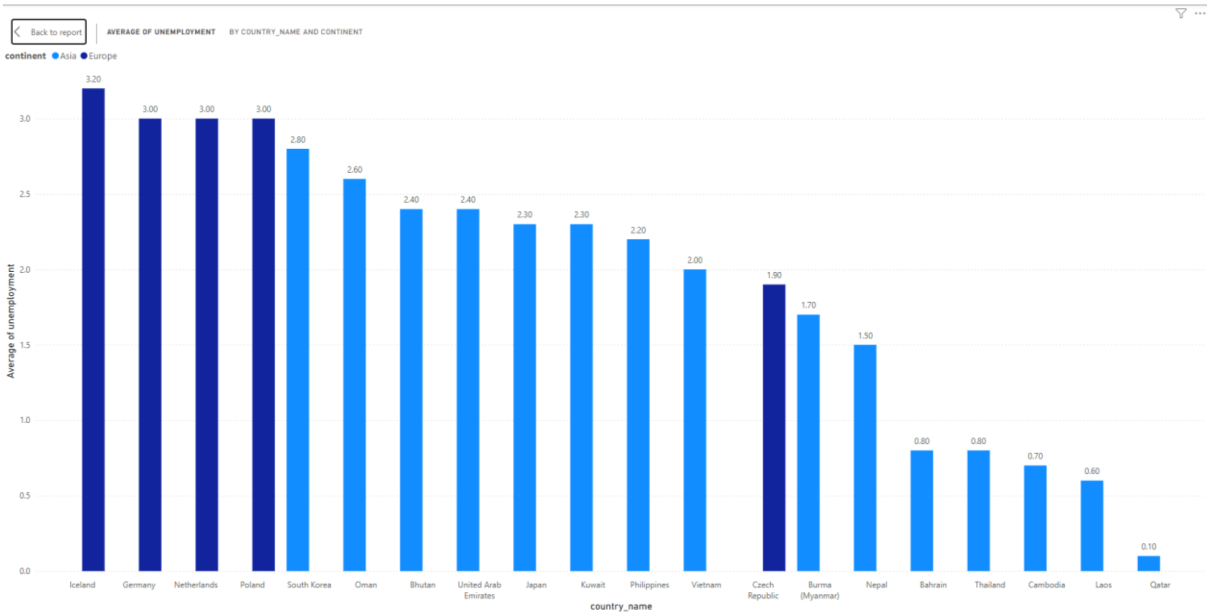

Figure 4: Average unemployment across the top countries in Europe and Asia, indicating that many Asian countries perform more strongly on labour market conditions, while European countries show greater variation.

Instead, the visual evidence suggests that the most attractive markets combine multiple strengths at once: reasonable market scale, strong development conditions, stable labour market outcomes, and meaningful trade connectivity. This interpretation aligns with the MABA/EDAS a frame work proposed (Hashemkhani Zolfani et al., 2021), which argues that international market selection is inherently a multiple-criteria decision problem requiring simultaneous consideration of several economic, social, political, and business factors.

The EDA therefore indicates that the most attractive markets are not the largest or fastest growing in isolation, but those that combine purchasing power, development quality, connectivity and sustainable demand conditions. Europe and Asia emerged as the most relevant regions for further consideration, but the country-level variation within both regions confirms that regional averages alone are insufficient for final market-entry decisions. This is consistent with the international market selection literature, emphasizing that better selection decisions come from structured evaluation of alternative markets and that poor selection imposes both direct and opportunity costs.

Part 1b: Predictive Data Analysis (PDA)

Predictive data analysis was used to test whether country performance could be estimated from the available indicators and to identify which factors contribute most strongly to those outcomes. The modelling stage followed a structured analytics process involving data cleaning, feature engineering, redundancy checks, target selection, and comparison of alternative models. This reflects the logic of CRISP-DM, which remains widely used as a framework for data science and predictive analytics, especially where business understanding, data understanding, and modelling need to be connected. Recent literature also shows that CRISP-DM is most effective when treated as an iterative and adaptable process rather than a rigid sequence of steps (Shimaoka et al., 2024).

The predictive stage examined three target variables: GDP growth, GDP per capita, and unemployment. These targets were selected because the exploratory analysis suggested that they represent different dimensions of country performance: short-term momentum, structural prosperity, and labour-market conditions. Multiple linear regression was used as the baseline interpretable model, while random forest was used as a non-linear comparison model. This model-comparison approach is consistent with predictive analytics studies in retail and data science, which show that different models can produce materially different predictive performance and that interpretable baselines remain useful even when stronger non-linear models are also tested.

The strongest predictive results were obtained for GDP per capita. In the report draft, the linear regression model explained approximately one-third of the variation in GDP per capita, while the random forest model explained substantially more, at around two-thirds (see Appendix C, Figures C12 to C17). This makes GDP per capita the strongest and most decision-relevant predictive target in the study. The stronger performance suggests that prosperity is more structurally linked to the available development, trade, and macroeconomic indicators than the other targets. In the linear model, life expectancy, business confidence, economic scale, and trade intensity were positively associated with GDP per capita, while unemployment was negatively associated with it (see Appendix C, Figures C13 and C14). In the random forest model, life expectancy emerged as the most important feature (see Appendix C, Figures C15 to C17). These results reinforce the EDA findings and are consistent with the broader literature on location attractiveness, which highlights human capital, development quality, and institutional readiness as key determinants of market attractiveness in the current environment.

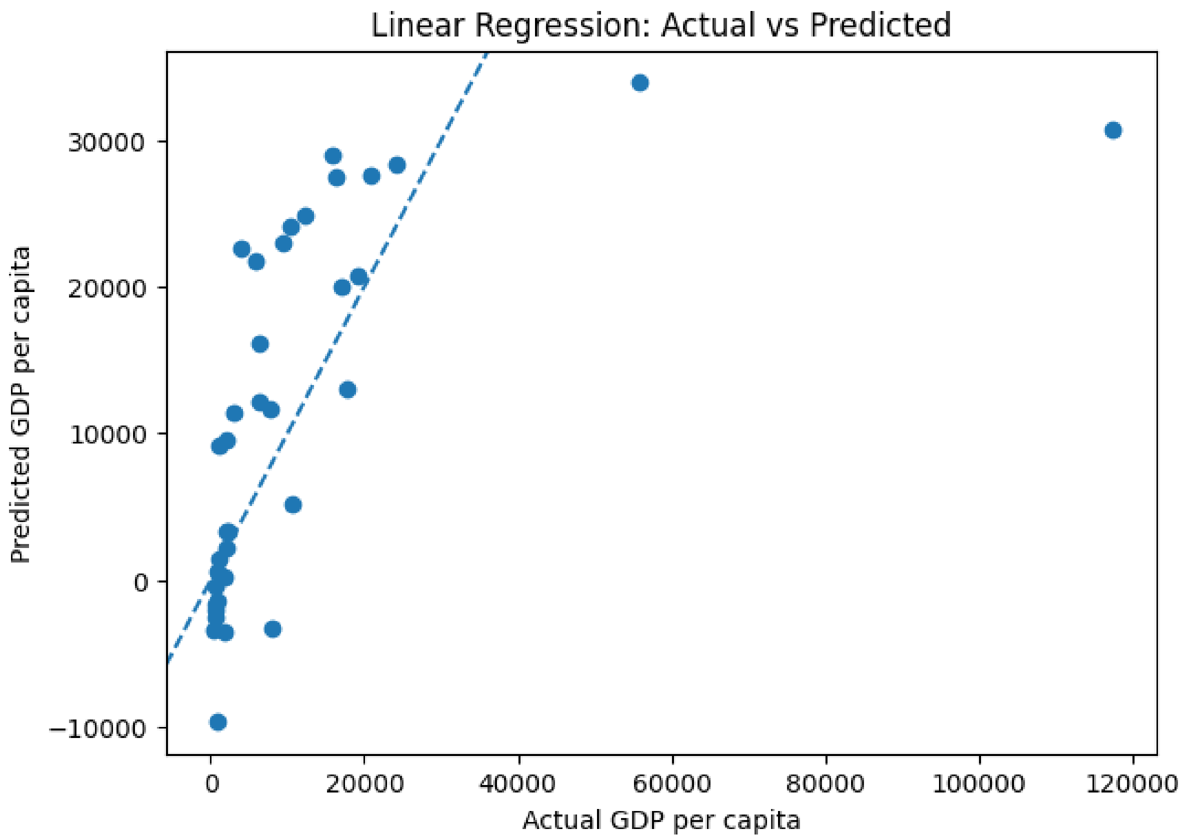

Figure 5: Predictive model performance for GDP per capita, showing that GDP per capita is the strongest and most reliable target in the analysis compared with the other outcomes tested.

Model Comparison

	Model	R2	MAE	RMSE
0	Linear Regression	0.343892	8999.979518	16831.831705
1	Random Forest	0.626276	5286.226487	12703.372301

Figure 6: Feature importance results for the GDP per capita model, indicating the relative contribution of the most influential predictors

By contrast, GDP growth proved difficult to predict. Both the linear regression and random forest models explained only a small share of the variation, and additional engineered features such as trade intensity and trade balance did not materially improve performance (see Appendix C, Figures C1 to C11). This suggests that GDP growth is influenced by factors not well captured in the current dataset, including macroeconomic shocks, policy differences, inflation, political events, and short-term volatility. For this reason, GDP growth is better treated as a contextual indicator rather than as the main predictive basis for market selection. This interpretation is consistent with CRISP-DM-oriented research showing that the usefulness of predictive analysis depends not only on modelling technique but also on whether the chosen target is structurally linked to the available data.

Unemployment also produced weak predictive performance. The random forest model explained only a limited share of the variation, while the linear model performed poorly (see Appendix C, Figures C18 to C23). This indicates that unemployment is not well explained by the available macroeconomic, trade, and demographic variables alone and likely depends on labour-market institutions, skills composition, regulation, and sectoral structure not included in the dataset. As a result, unemployment is more useful in this study as a contextual risk indicator within the EDA than as a robust predictive target. This is also in line with recent AI and analytics process literature, which stresses that modelling should be assessed in terms of practical usefulness and data adequacy, not just whether an algorithm can technically be applied.

Taken together, the predictive analysis shows that the usefulness of PDA depends heavily on the target selected. GDP per capita was the strongest and most useful predictive outcome because it was structurally related to the available indicators and could therefore support more reliable business interpretation. GDP growth and unemployment, although weaker as predictive targets, were still informative because they highlighted the limits of the dataset and showed that not all aspects of country performance can be estimated equally well from the variables available. This is consistent with both predictive analytics and CRISP-DM research, which emphasises that model value comes from choosing relevant targets, comparing models appropriately, and interpreting results within a business decision context.

Part 2: Discussion

This report aimed to identify which countries appear most expensive for international retail expansion using country-level quantitative analysis. Overall, the findings show that market attractiveness is multidimensional and should not be judged using a single measure such as GDP, GDP growth, or trade balance alone. Instead, the strongest markets are those that combine economic scale, purchasing power, development quality, labour market stability, and trade connectivity. This is consistent with the international market selection literature, which argues that foreign market choice is complex, multicriteria decision rather than a simple ranking exercise (Hashemkhani Zolfani et al., 2021, Francioni and Martín Martín, 2024). The EDA provided the clearest strategic picture. At regional level, Europe and Asia emerged as the strongest regions, but for different reasons. Europe performed more well on the development indicators, indicating a stable and low risk operating environment, while Asia performed stronger on trade activity, net exports, indicating stronger scale and long-term growth potential (see Appendix A, Figures A1, A2, A4, A10, A11, A12 and A15). This supports the OECD view that attractiveness should be understood as a profile of strengths and weaknesses rather than a single score (OECD, 2024).

At a country level, the analysis showed that regional averages hide important variation. China, Germany and Japan stood out for economic and trade scale, while Singapore, Iceland, Finland appeared stronger on development and labour-related indicators. Bahrain, Oman, and Qatar showed stronger demographic growth (see Appendix B, Figures B5 to B10). This suggests that size alone is not enough to determine attractiveness. Instead, the strongest candidate markets are those that combine scale with strong operating conditions. This is also consistent with retail internationalisation research, which shows that attractive markets differ even within the same broad region (Mladenović et al., 2020). The PDA added a second layer of insight by showing which outcomes could be predicated more reliably from the dataset. The most important result was that GDP per capita was much more predictable than either GDP growth or unemployment. The GDP per capita models performed relatively well, the random forest performed was good, suggesting that structural prosperity is more closely linked to the available indicators than short-term growth or labour-market outcomes (see Appendix C, Figures C12 to C17). This reinforces the EDA finding that development quality and purchasing power are central to market attractiveness. It also supports recent location attractiveness research, which highlights human capital, development quality, and institutional readiness as important drivers of location strength (Andy et al., 2024).

By contrast, GDP growth and unemployment were difficult to predict (ee Appendix C, Figures C1 to C11 and C18 to C23). This suggests that these variables depend on additional factors not captured in the dataset, such as inflation, policy, regulation, and labour market structure. Rather than making the models useless, this highlights the limits of the available data and supports the CRISP-DM view that predictive models should be judged by their practical usefulness and fit with the business problem, not just by whether a model can be produced (Bokrantz et al., 2024, Shimaoka et al., 2024).

The EDA and PDA work together. The EDA identifies where the strongest regional and country-level opportunities appear to be, while the PDA shows that GDP per capita is the most useful predictive indicator for market evaluation in this dataset. For managers, the key implication is that Europe appears strongest for lower-risk, development-led entry conditions, while Asia appears strongest for scale, trade connectivity, and longer-term growth potential. However, final expansion decisions should still be made at country level, and should be treated as part of a broader screening process rather than as a final answer on their own (Francioni and Martín Martín, 2024, Intern-report, 2026)

Based on the combined EDA and PDA, the report should recommend a two-stage market screening approach rather than an immediate final market-entry choice. The evidence suggests that Europe and Asia should be prioritised, but for different strategic reasons. Europe appears more suitable for lower-risk expansion because it performs stronger on wealth and development indicators, while Asia looks suitable for scale-led and longer-term growth strategies because it performs stronger on trade activity, net exports, and lagged growth (see Appendix A, Figures A1, A2, A4, A10, A11, A12 and A15). This is consistent with the literature, which argues that market selection should reflect multiple criteria and that different markets may be attractive for different strategic purposes (Hashemkhani Zolfani et al., 2021, Francioni and Martín Martín, 2024, Intern-report, 2026)

The first recommendation is therefore to prioritise Europe for stable, lower-risk market entry. The region's stronger GDP per capita, life expectancy, and lower infant mortality suggest more mature and resilient operating conditions, which are likely to support more predictable retail demand and lower execution risk. Within Europe, the next step should be to shortlist countries that combine purchasing power with favourable labour and development conditions, rather than relying on size alone.

The second recommendation is to prioritise selected Asian markets for scale and future growth potential. Asia's strength in trade activity, connectivity, and lagged growth suggests that it offers the strongest long-term expansion potential, but the country-level analysis shows that not all Asian markets are equally attractive. As a result, managers should focus on countries that combine trade connectivity and market scale with acceptable development conditions and business confidence, rather than treating Asia as a single opportunity set.

A third recommendation is to place greater weight on GDP per capita and development-related indicators than on GDP growth when evaluating markets. The predictive analysis showed that GDP per capita was the strongest and most reliable target, whereas GDP growth and unemployment were difficult to predict with the available dataset (see Appendix C, Figures C1 to C23, especially C12 to C17). This suggests that structurally stable indicators such as purchasing power, life expectancy, and development quality are more decision-useful for market screening than short-term growth rates alone.

Finally, the report should recommend a second-stage qualitative review before any final entry decision is made. The current analysis provides a strong screening framework, but it does not fully capture factors such as regulation, competition, digital readiness, supply-chain conditions, or local partnership opportunities. Consistent with the intern brief and the market selection literature, the quantitative findings should therefore be used to narrow options for managers, not replace managerial judgement.

Conclusion

This report shows that market attractiveness for international retail expansion is best understood as a multi-dimensional concept rather than a single economic measure. The EDA found that Europe and Asia are the strongest regions overall, but for different reasons: Europe appears stronger on development quality and lower-risk operating conditions, while Asia appears stronger on trade connectivity, scale, and longer-term growth potential (see Appendix A). At country level, the analysis also showed that regional averages conceal important differences, meaning final decisions must be made at country level rather than through broad regional comparisons alone.

The PDA strengthened this conclusion by showing that GDP per capita was the most useful predictive target in the dataset, while GDP growth and unemployment were much harder to model reliably (see Appendix C). This suggests that structurally stable indicators such as purchasing power and development quality are more useful for market screening than short-term growth measures alone. Overall, the report provides a practical screening framework for managers: prioritise Europe for lower-risk entry conditions, prioritise selected Asian markets for scale and long-term growth potential, and use the results as the basis for a more detailed second-stage commercial assessment rather than as a final decision on their own.

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Appendix

Appendix A: Continent level EDA

GDP/Capita vs Life Expectancy

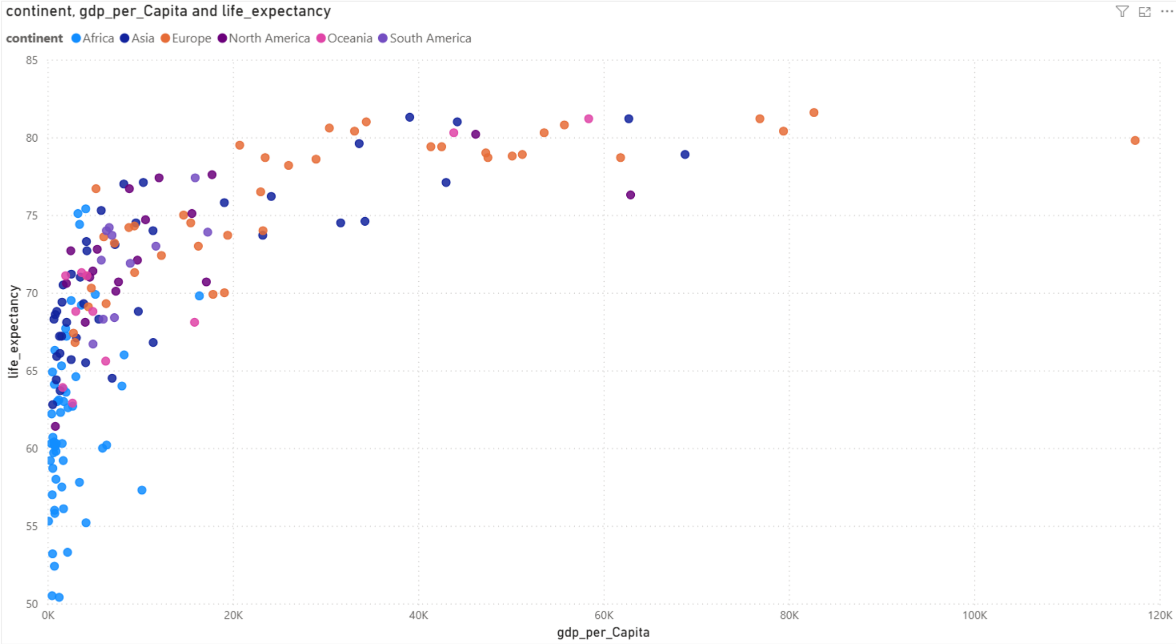

Figure A 1

GDP per Capita vs Infant mortality

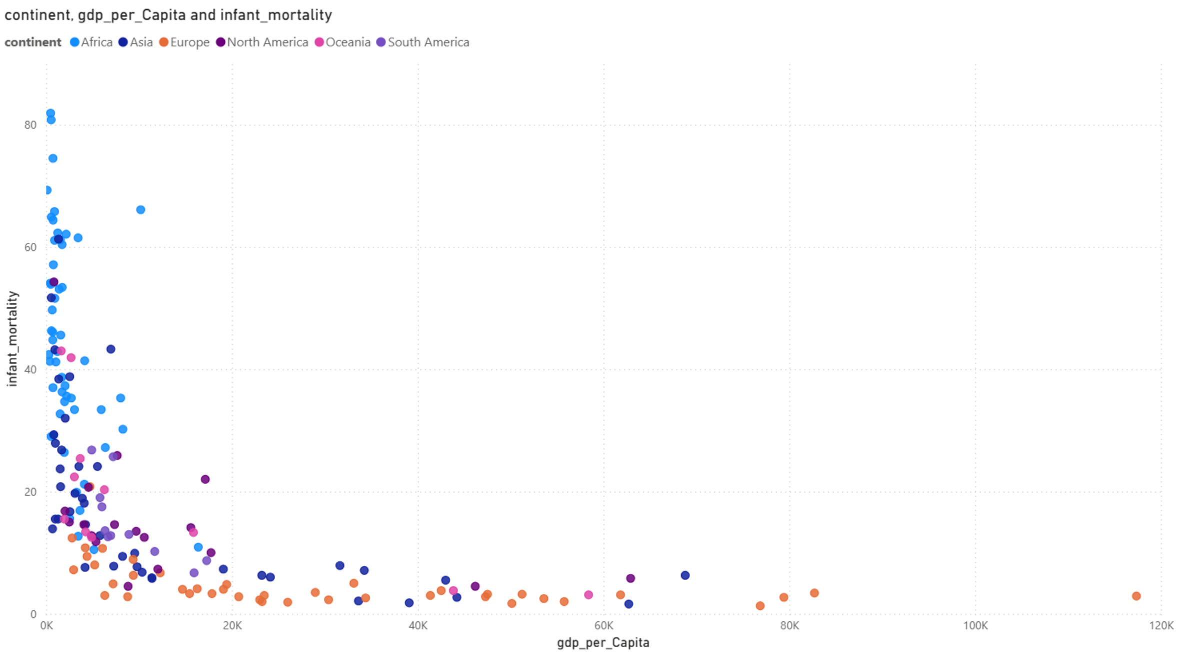

Figure A 2

GDP Growth vs GDP per Capita

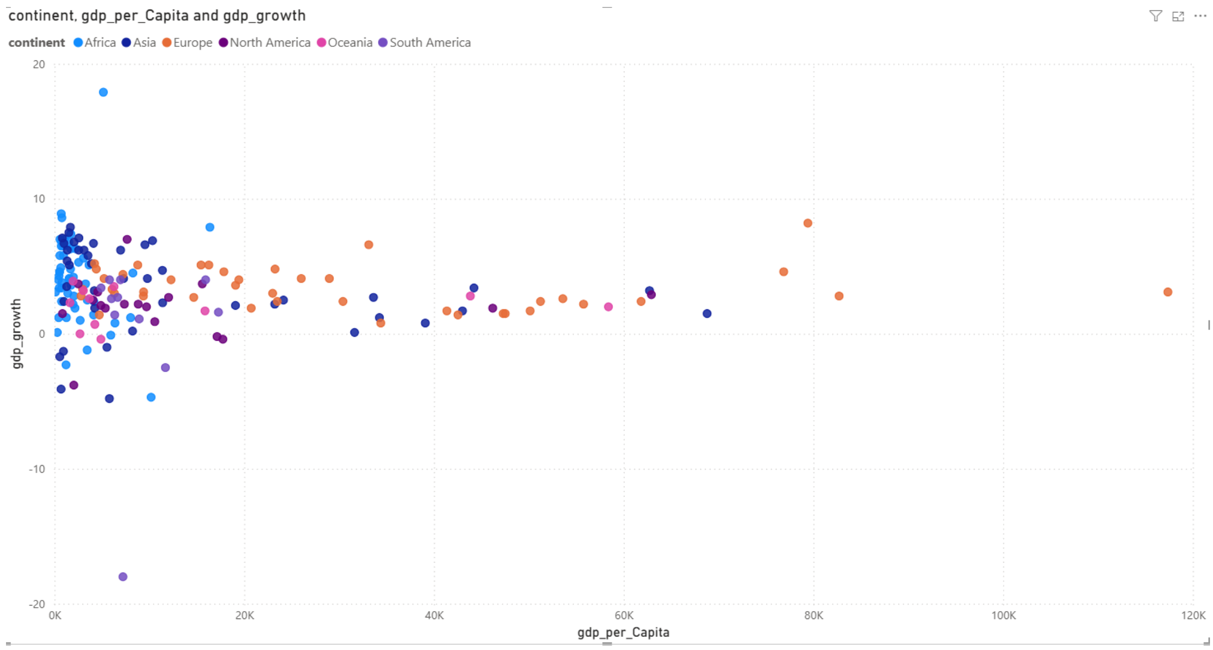

Figure A 3

Population Growth vs Urban Population Growth

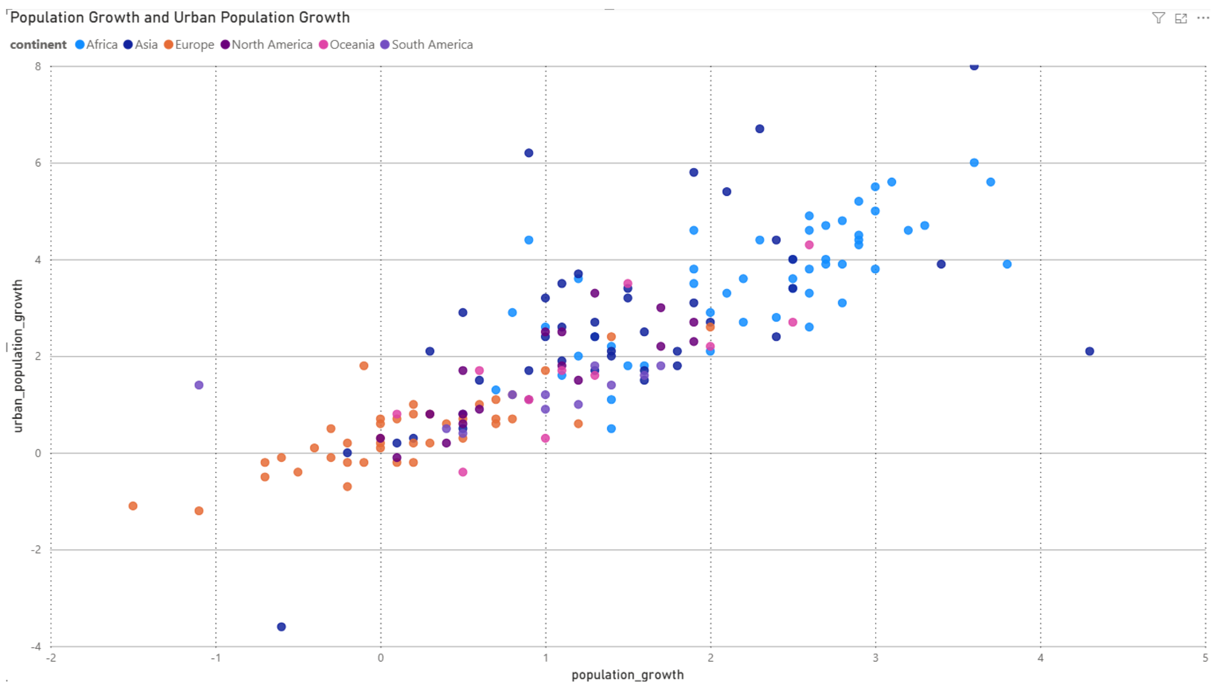

Figure A 4

Imports vs Exports

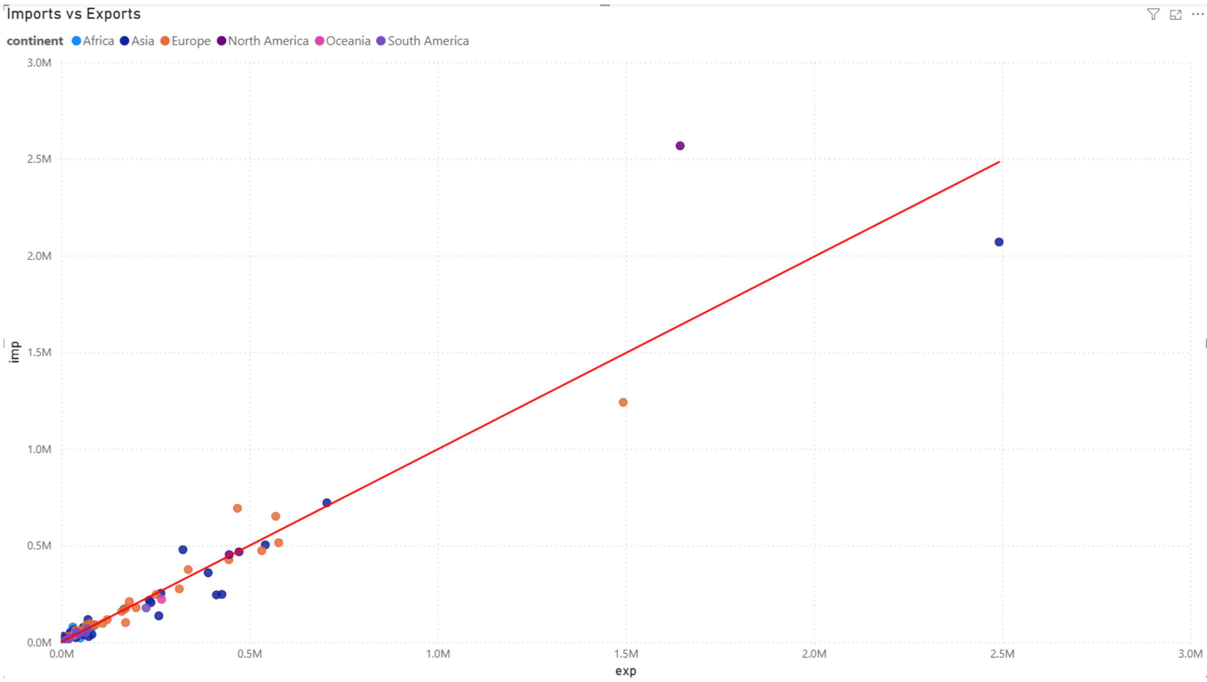

Figure A 5

Correlation Matrix

	GDP	Life Expectancy	Unemployment	Imports	Urban Population Growth	Exports	Infant Mortality	GDP Growth	Population Growth	GDP Per Capita
GDP	1.00									
Life Expectancy	0.21	1.00								
Unemployment	-0.08	-0.11	1.00							
Imports	0.95	0.31	-0.12	1.00						
Urban Population Growth	-0.11	-0.49	-0.06	-0.16	1.00					
Exports	0.86	0.32	-0.14	0.96	-0.16	1.00				
Infant Mortality	-0.17	-0.91	0.05	-0.26	0.60	-0.27	1.00			
GDP Growth	0.00	-0.09	-0.18	-0.01	0.07	-0.01	0.07	1.00		
Population Growth	-0.15	-0.52	-0.12	-0.22	0.83	-0.22	0.63	0.17	1.00	
GDP Per Capita	0.26	0.67	-0.17	0.36	-0.27	0.36	-0.53	-0.09	-0.26	1.00

Figure A 6

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Boxplot

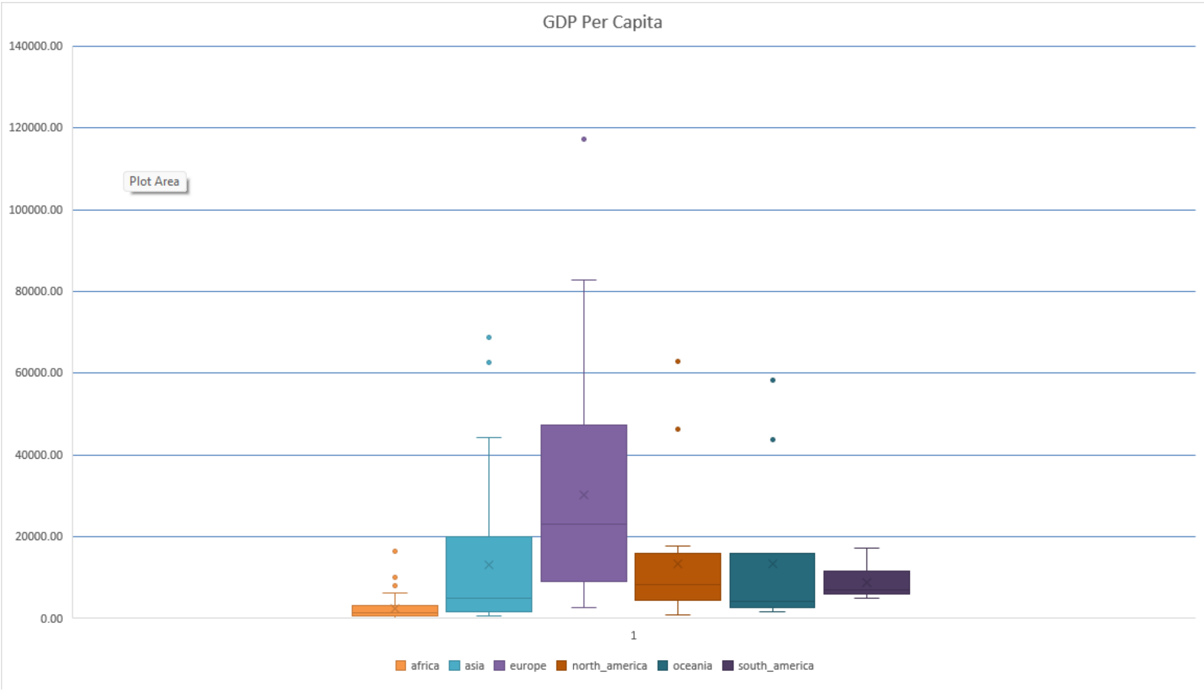


Figure A 7

Average of population_growth and Average of urban_population_growth by continent

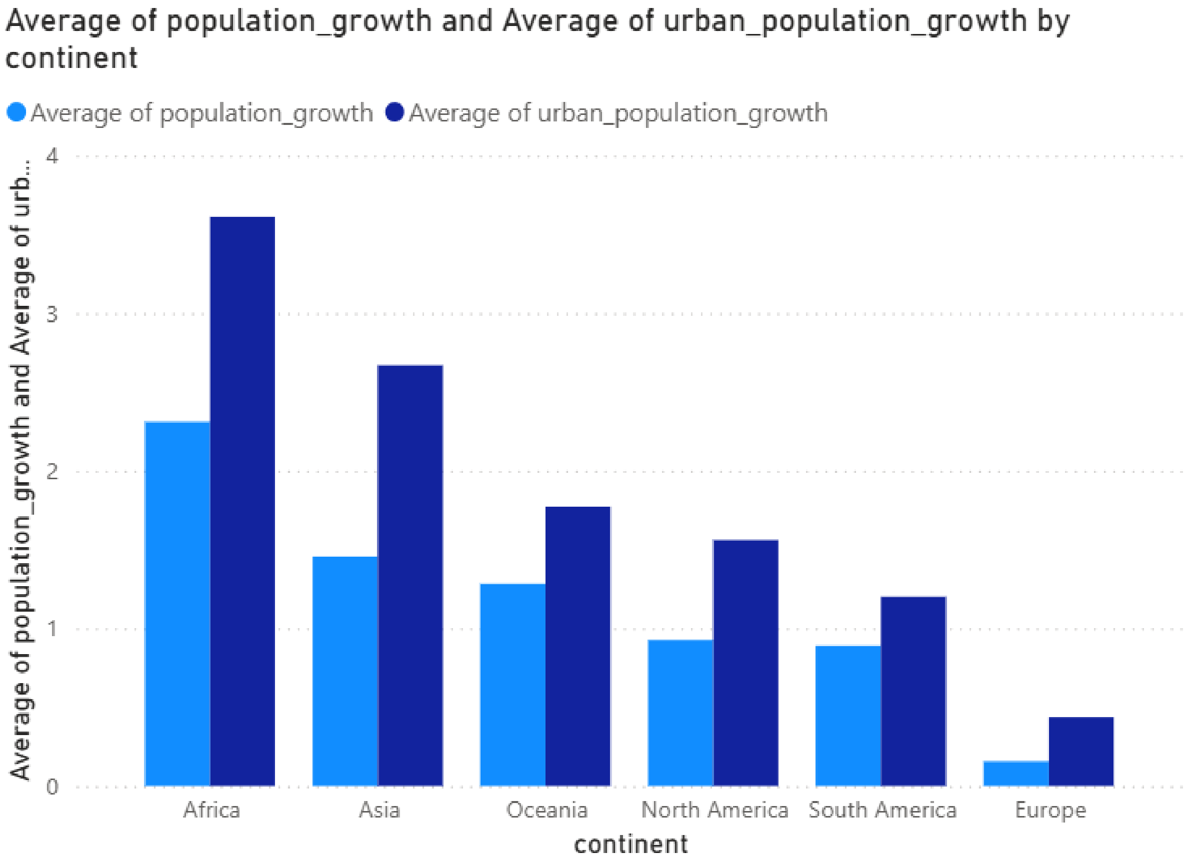


Figure A 8

Urbanisation_Gap by continent

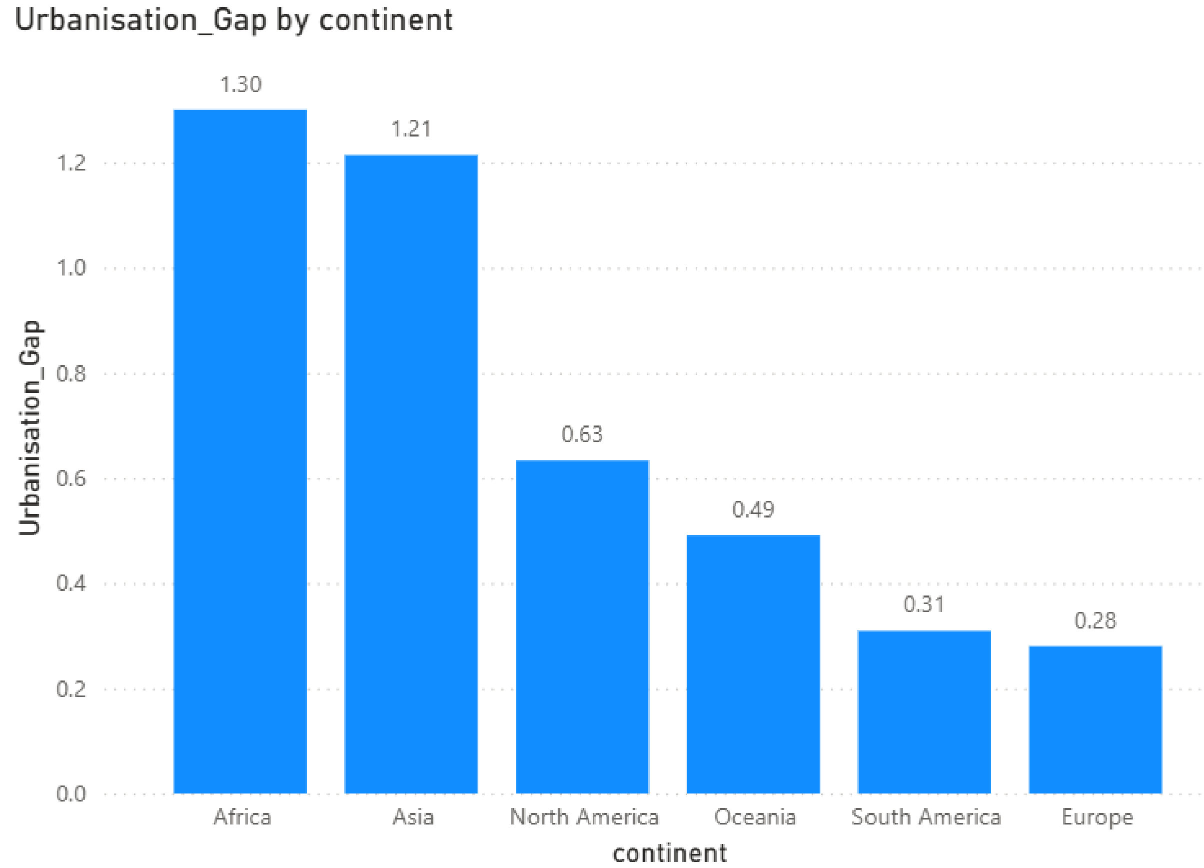


Figure A 9

Average of imp and Average of exp by continent

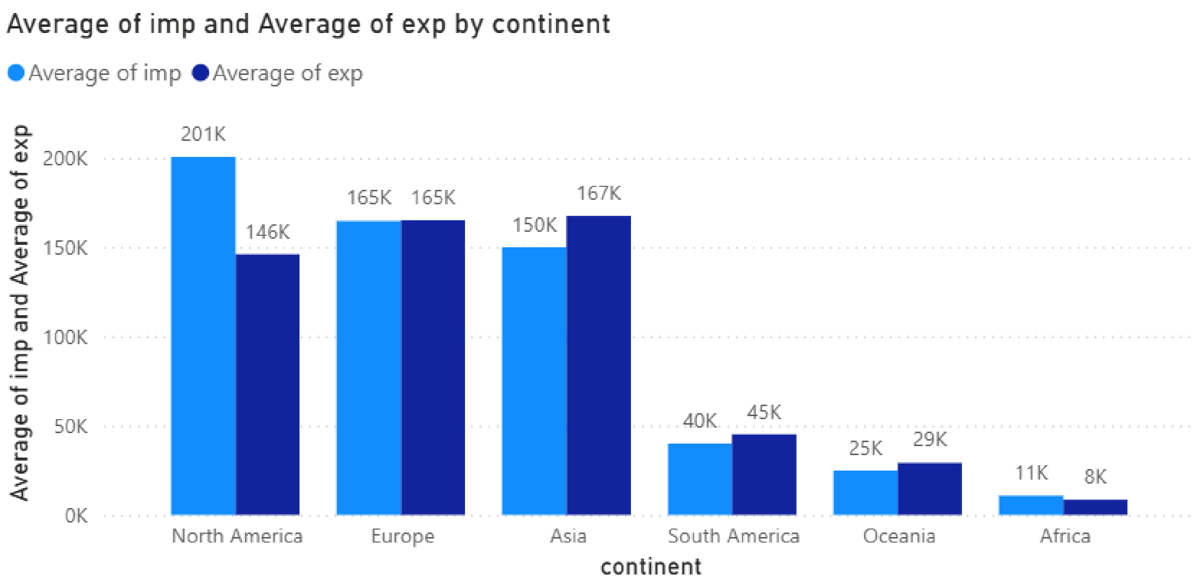


Figure A 10

Net Exports by continent

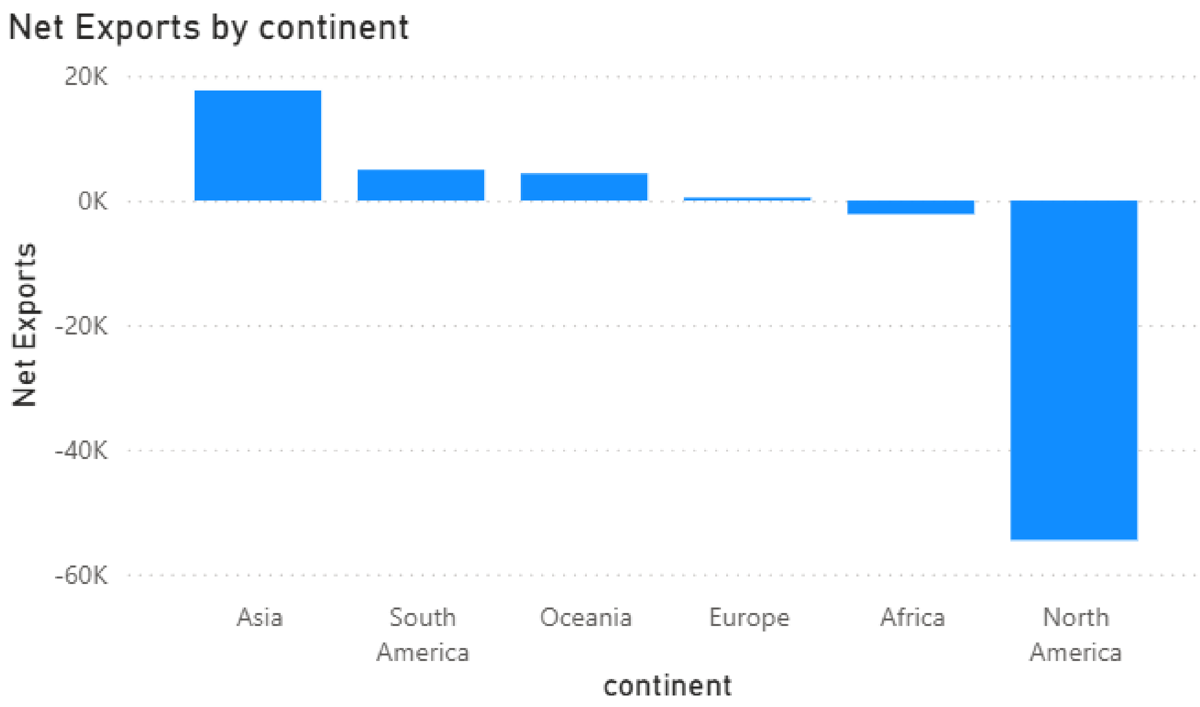


Figure A 11

Average GDP/Capita

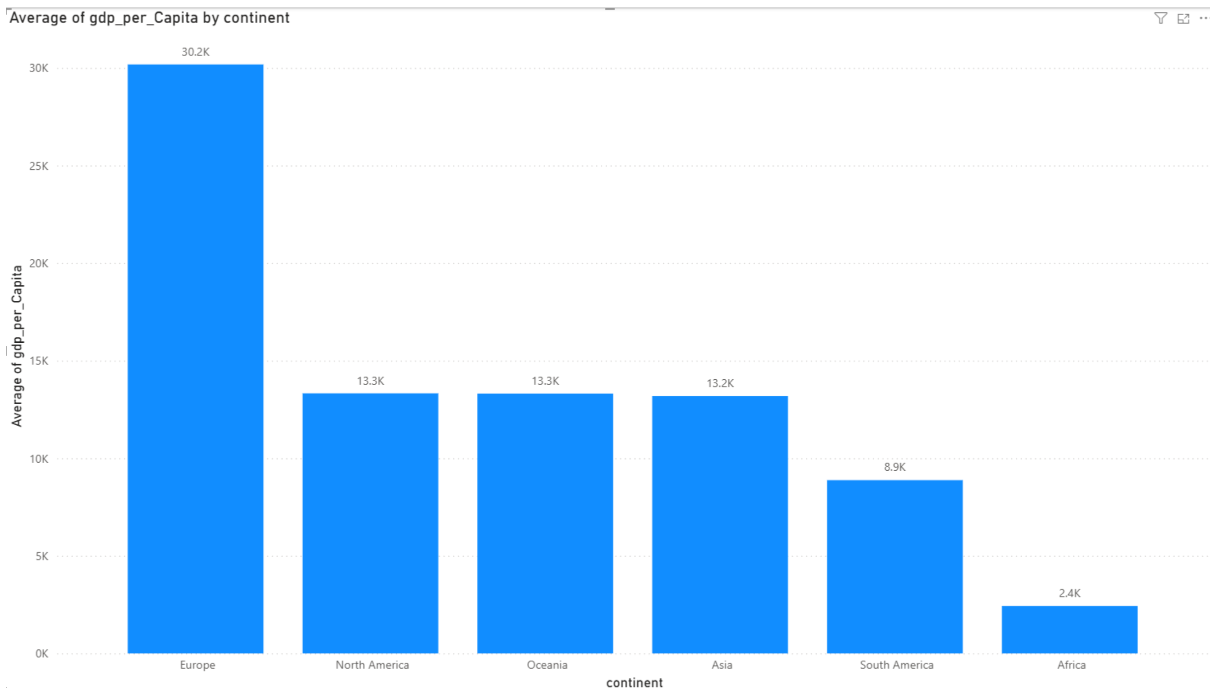


Figure A 12

Average of unemployment by continent

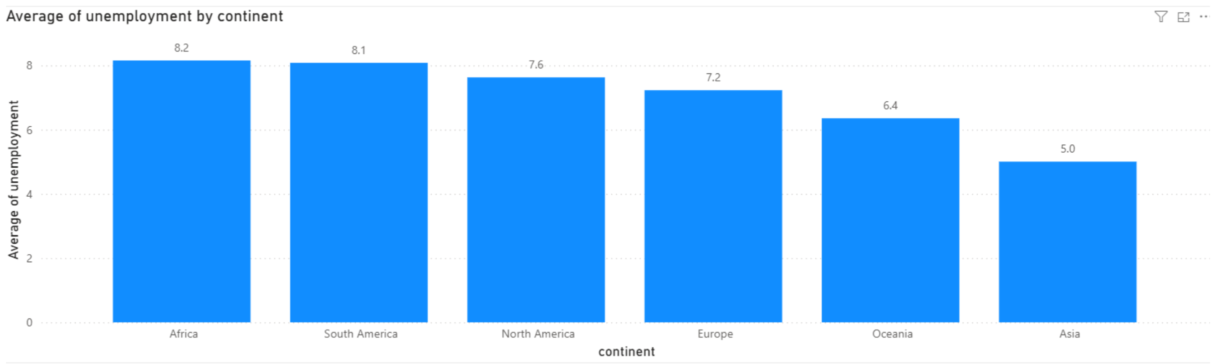


Figure A 13

Average of infant mortality by continent

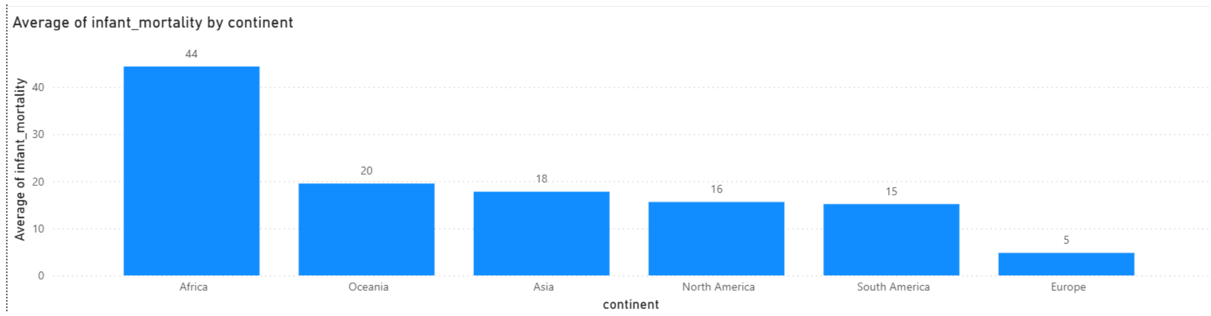


Figure A 14

Trade Activity by continent

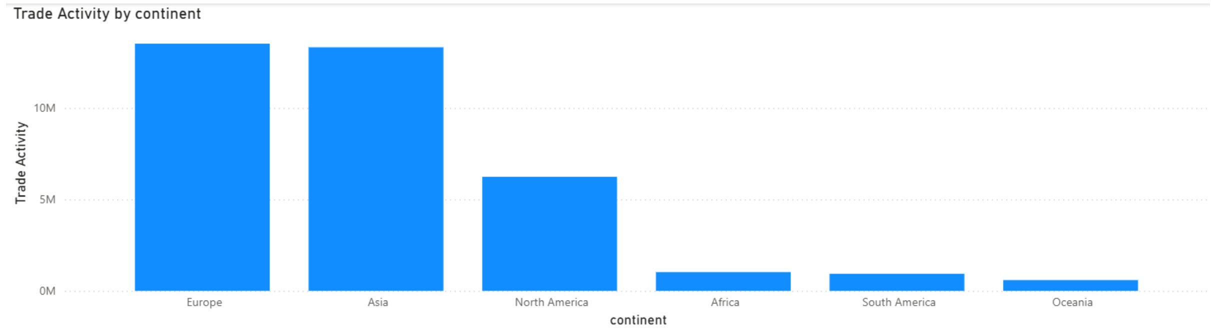


Figure A 15

Trade Intensity by continent

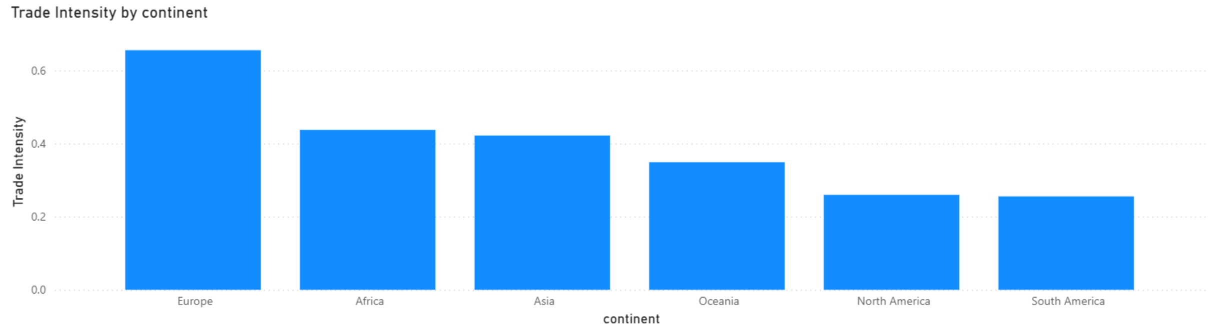
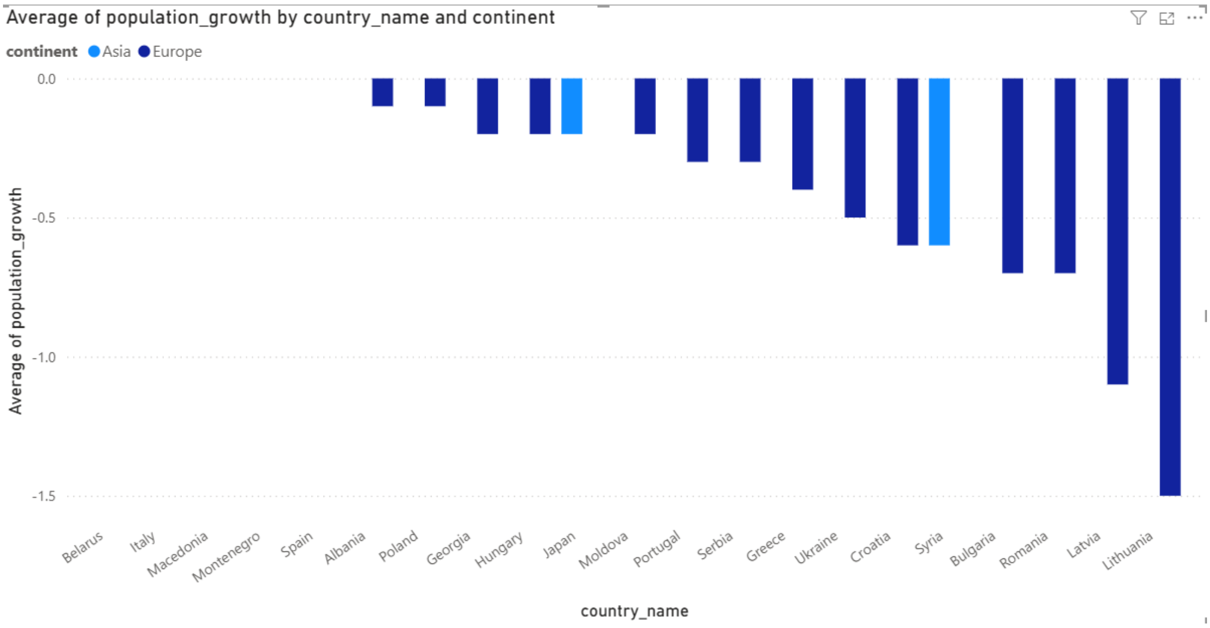
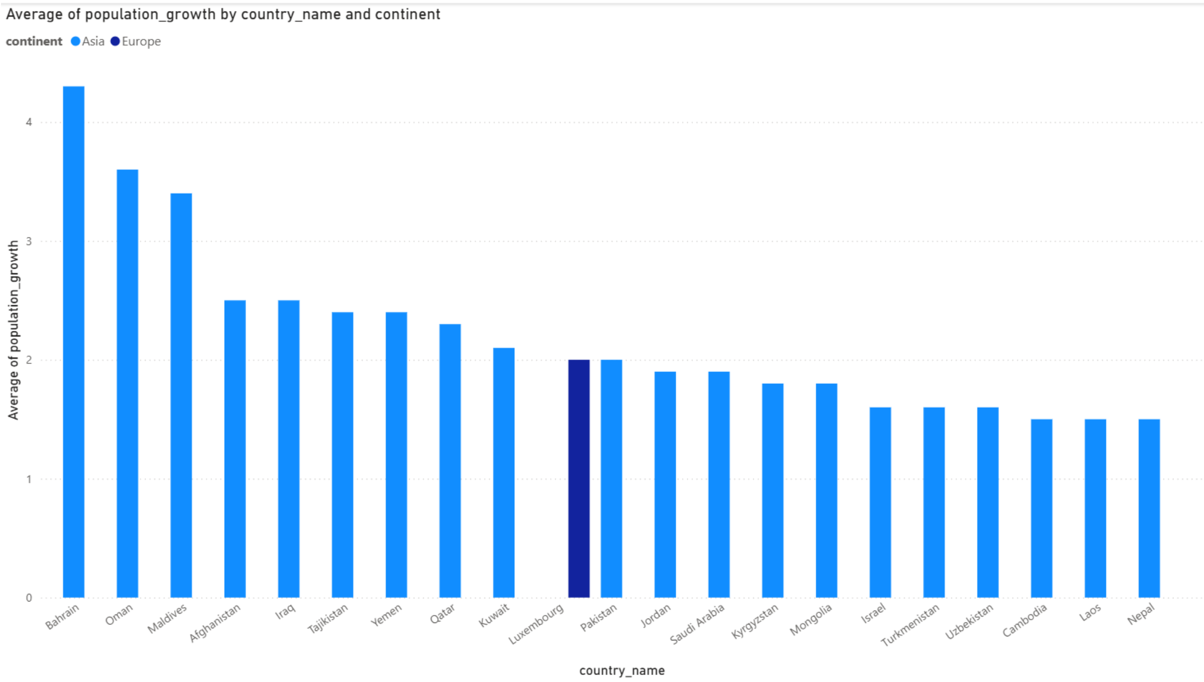


Figure A 16

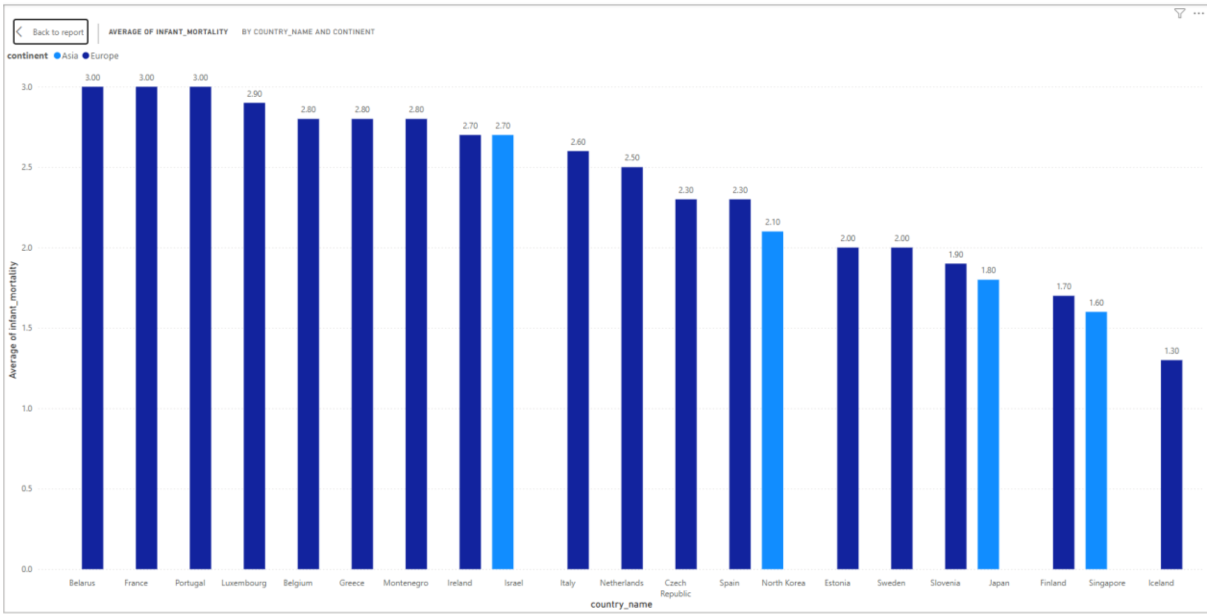
Appendix B: Country Level EDA



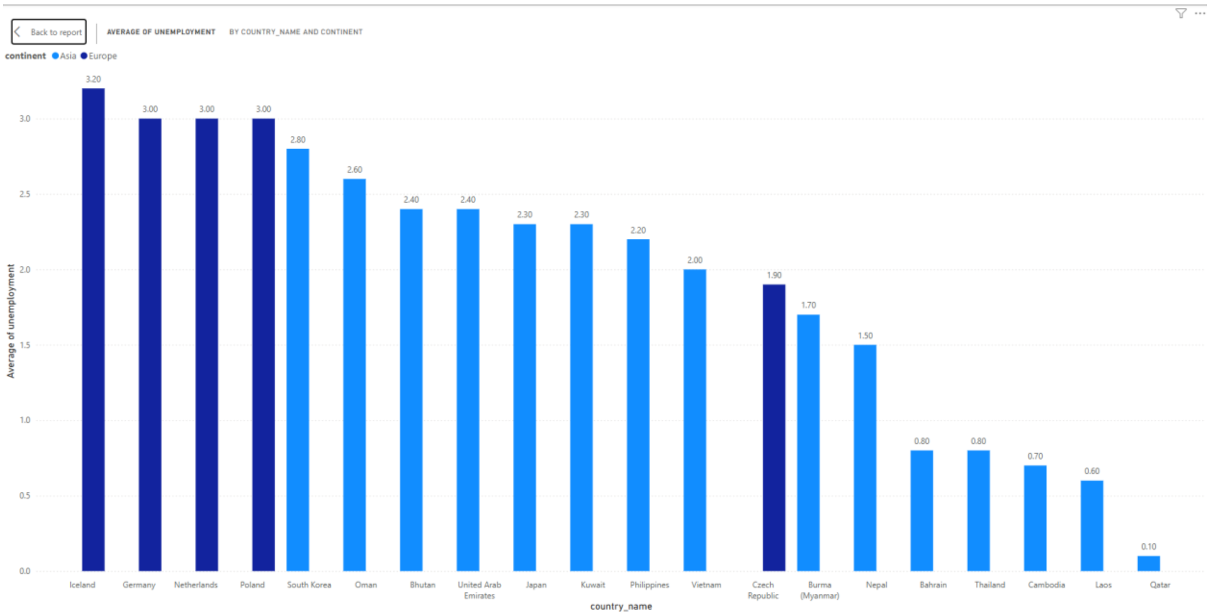
Appendix B: Figure 1



Appendix B: Figure 2



Appendix B: Figure 3



Appendix B: Figure 4

This visual is assessing the 20 top countries in Europe and Asia for the average unemployment. In this plot it shows that the top 10 with low unemployment are mostly in Asia while most of the countries in

Appendix C: Predictive Analysis

Model A: GDP Growth

Model Comparison

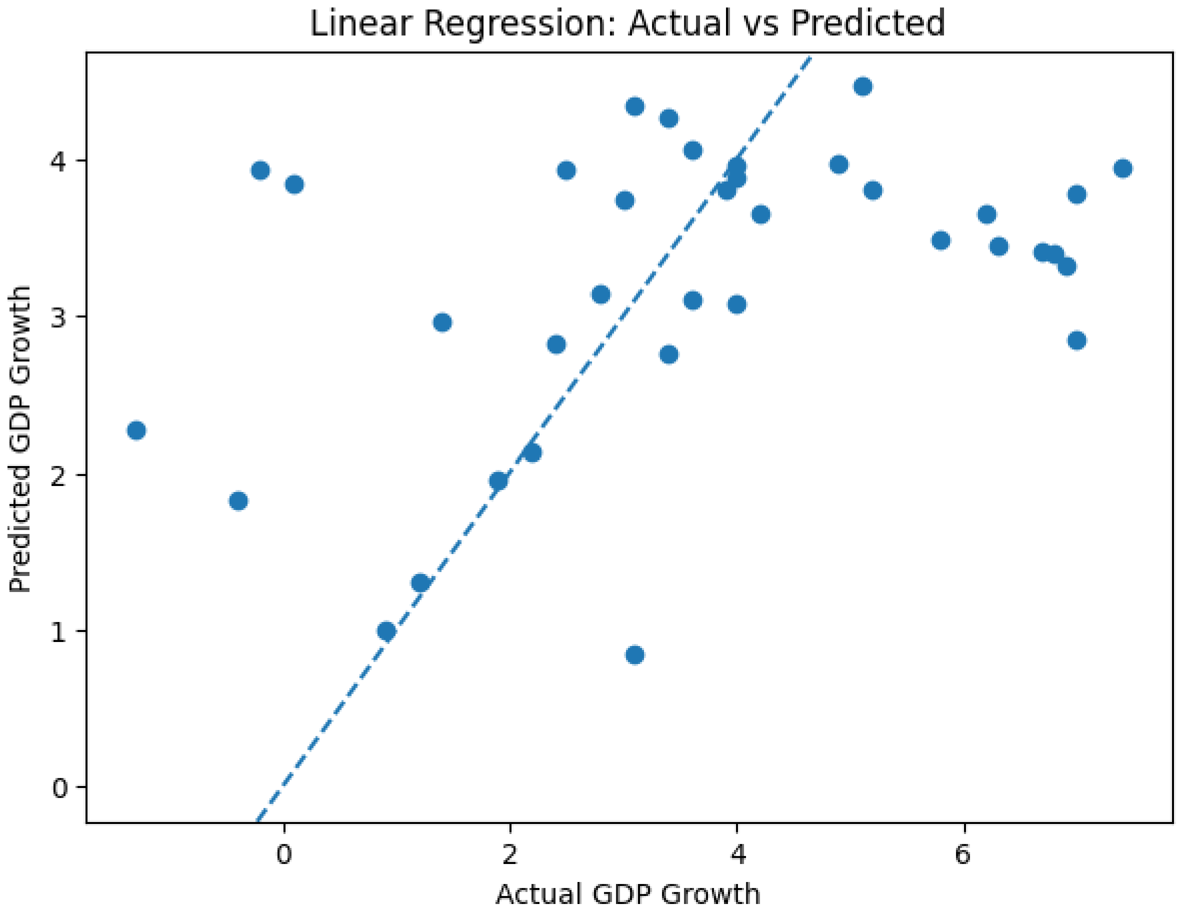
	Model	R2	MAE	RMSE
0	Linear Regression	0.141666	1.608868	2.117825
1	Random Forest	0.145299	1.662149	2.113339

Appendix C: Figure 1

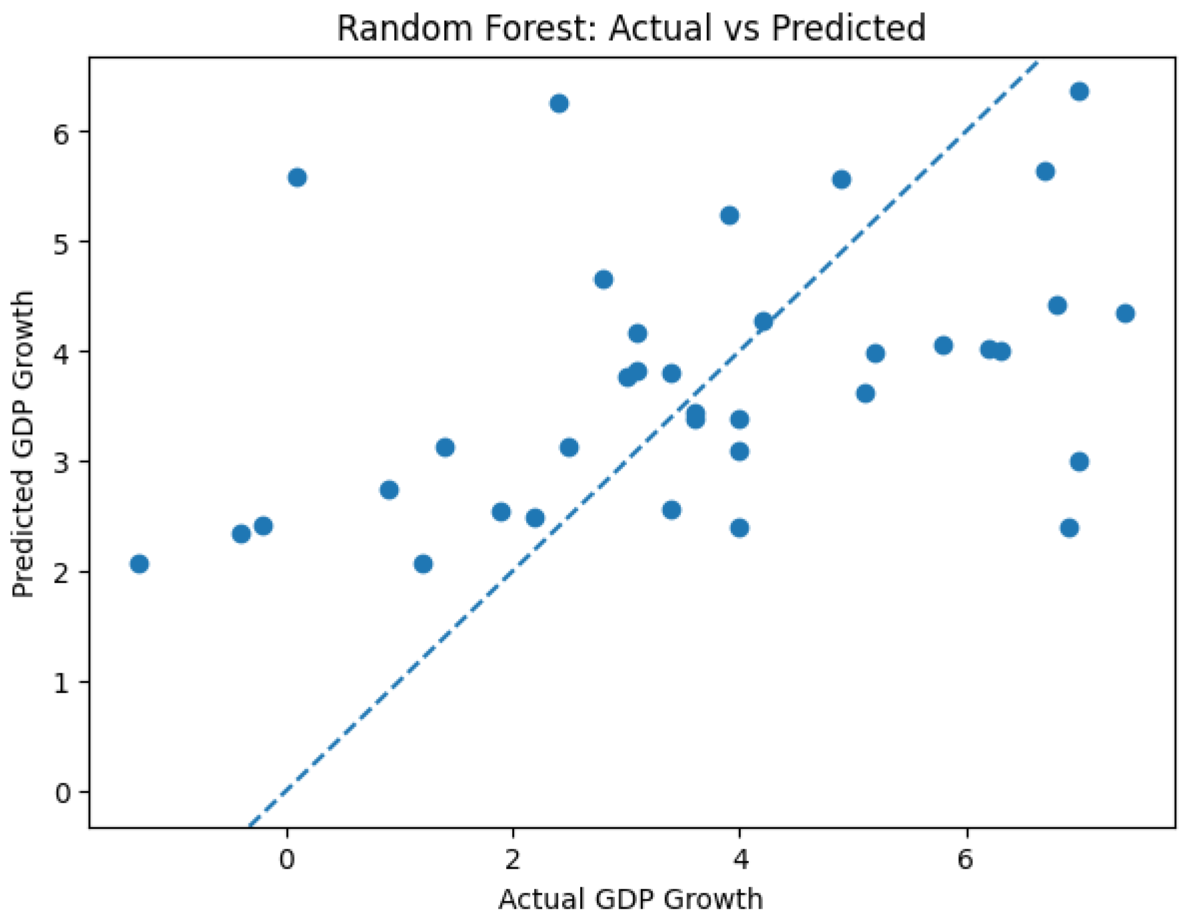
Linear Regression Coefficients

	Feature	Coefficient
5	business_confidence_index	0.374549
4	trade_intensity	0.350726
2	urban_population_growth	-0.038133
3	life_expectancy	-0.306345
0	gdp_per_Capita	-0.445687
1	unemployment	-0.621917

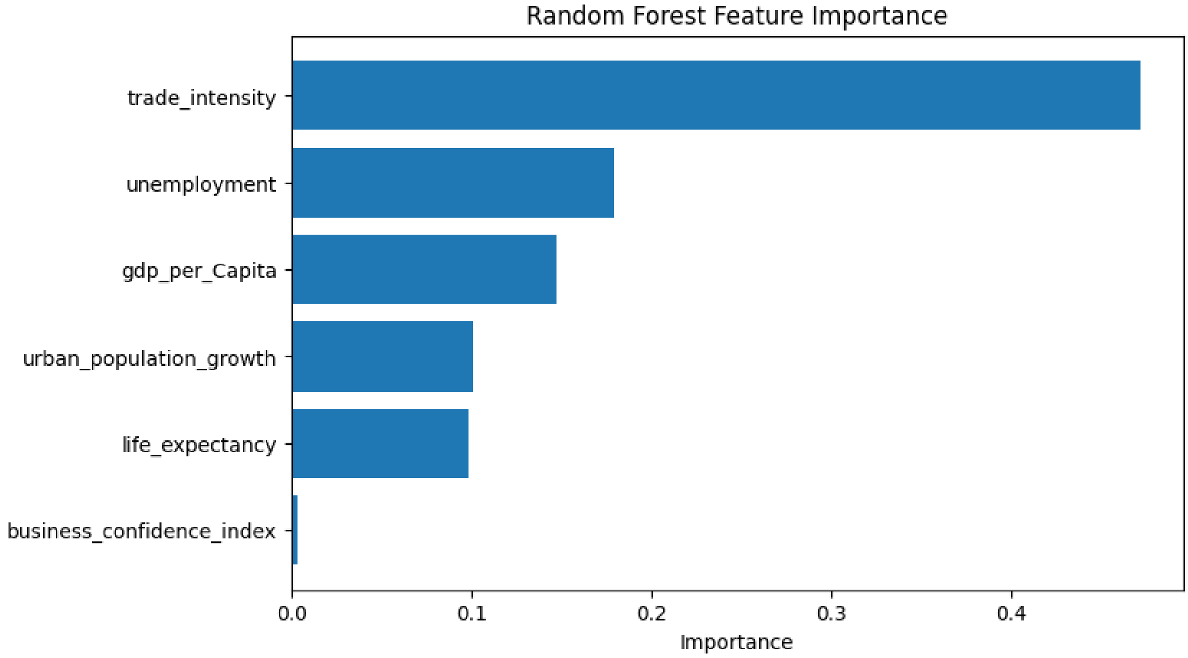
Appendix C: Figure 2



Appendix C: Figure 3



Appendix C: Figure 4



Appendix C: Figure 5

Improved Model with added features

```
Model Comparison
               Model        R2       MAE      RMSE
0  Linear Regression  0.135480  1.536331  2.125444
1      Random Forest  0.133673  1.621822  2.127664
```

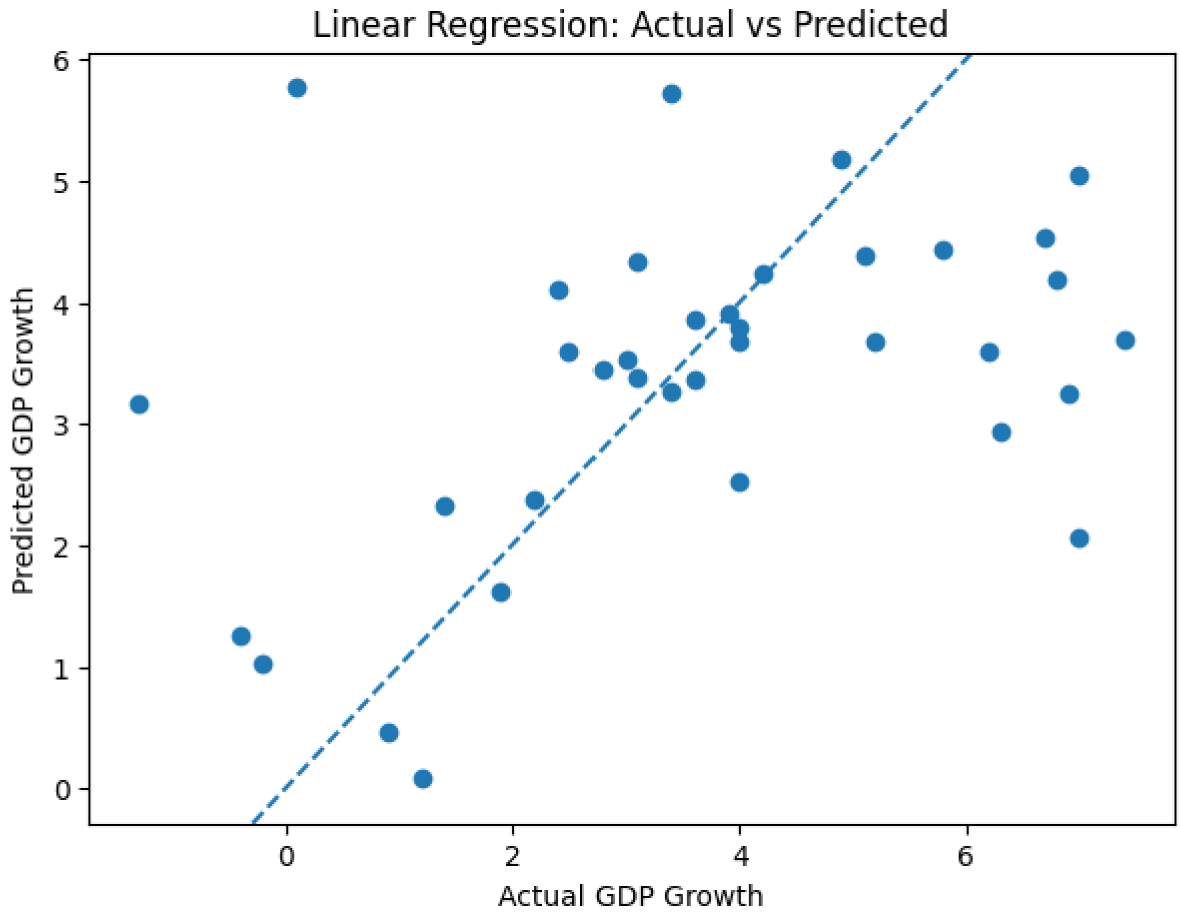
Appendix C: Figure 6

```
Linear Regression Coefficients
                    Feature  Coefficient
4           trade_intensity     0.608972
9  business_confidence_index    0.582727
0            gdp_per_Capita     0.515442
5             trade_balance     0.007877
3           life_expectancy    -0.005657
8                   log_gdp    -0.044699
2    urban_population_growth   -0.173339
1              unemployment    -0.653396
6         trade_balance_pct    -0.740463
7        log_gdp_per_Capita    -1.588748
```

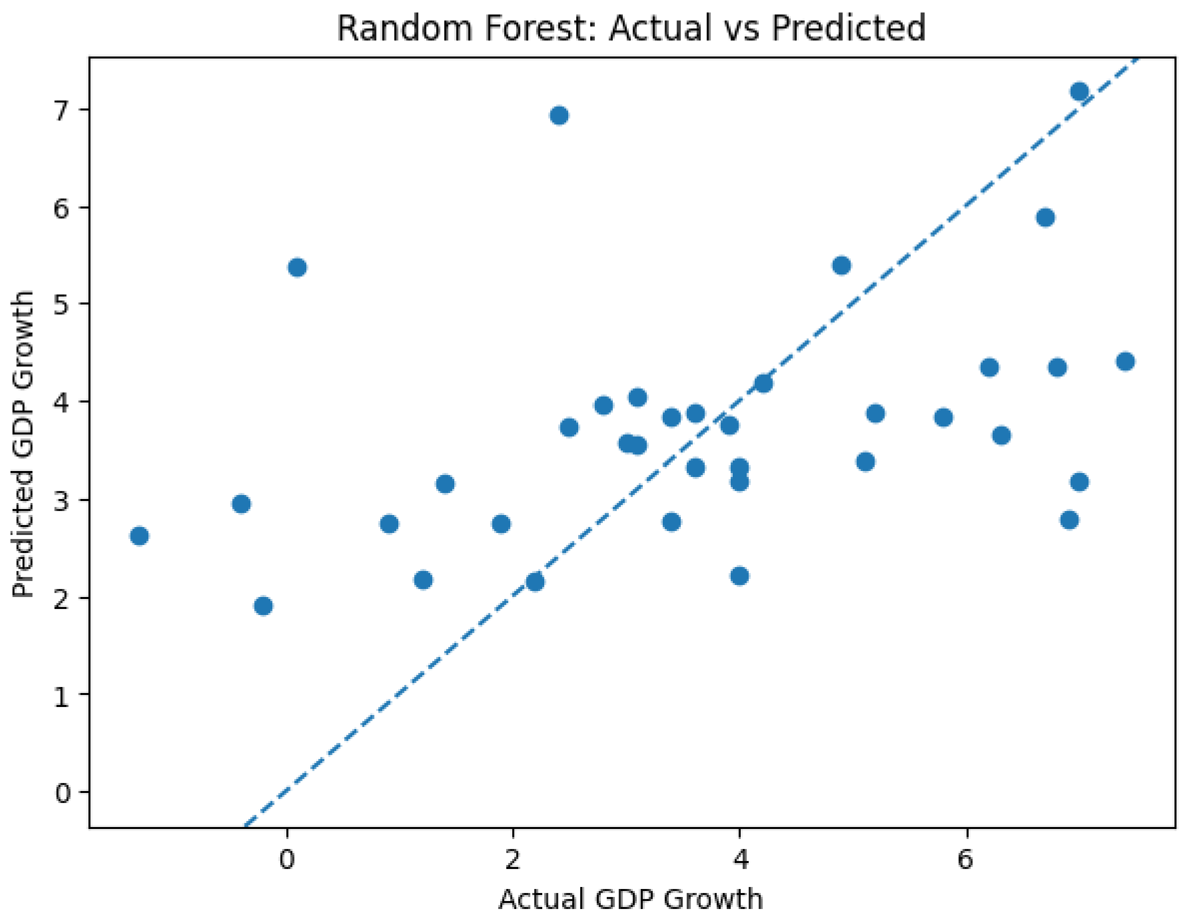
Appendix C: Figure 7

```
Random Forest Feature Importances
                    Feature  Importance
4           trade_intensity    0.429170
1              unemployment    0.151277
2    urban_population_growth   0.079282
6         trade_balance_pct    0.064528
0            gdp_per_Capita    0.063866
7        log_gdp_per_Capita    0.061890
3           life_expectancy    0.059515
8                   log_gdp    0.047512
5             trade_balance    0.040988
9  business_confidence_index   0.001972
```

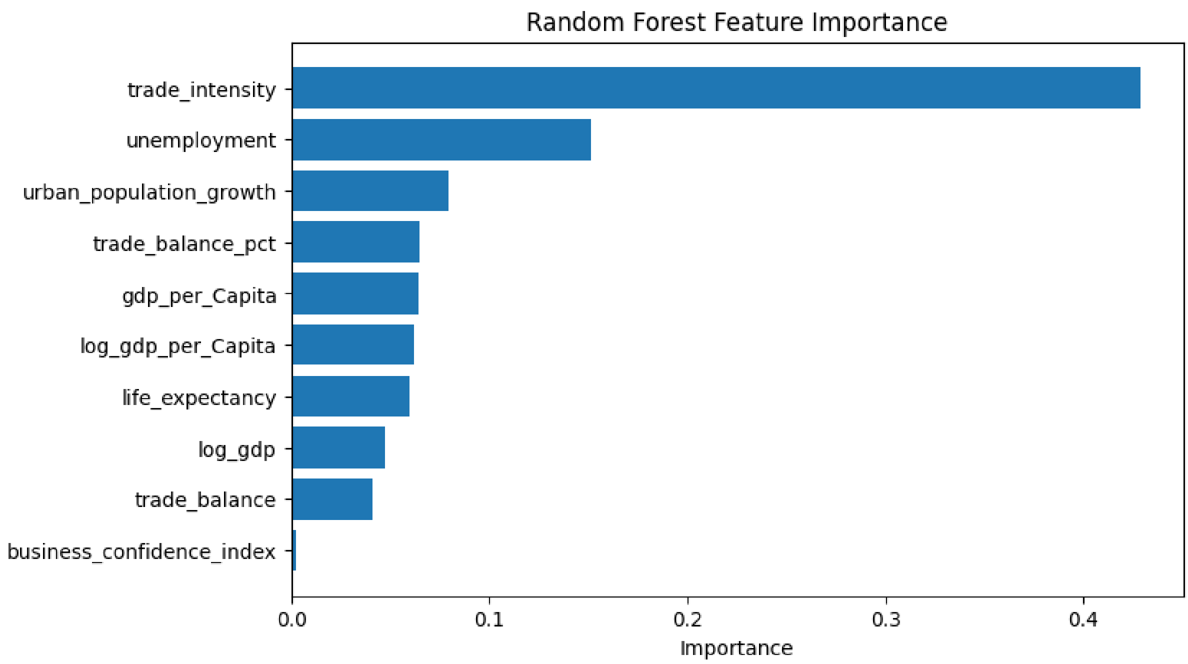
Appendix C: Figure 8



Appendix C: Figure 9



Appendix C: Figure 10



Appendix C: Figure 11

Model B: GDP per Capita

Target Variable = GDP per capita

Model Comparison

	Model	R2	MAE	RMSE
0	Linear Regression	0.343892	8999.979518	16831.831705
1	Random Forest	0.626276	5286.226487	12703.372301

Appendix C: Figure 12

Linear Regression Coefficients

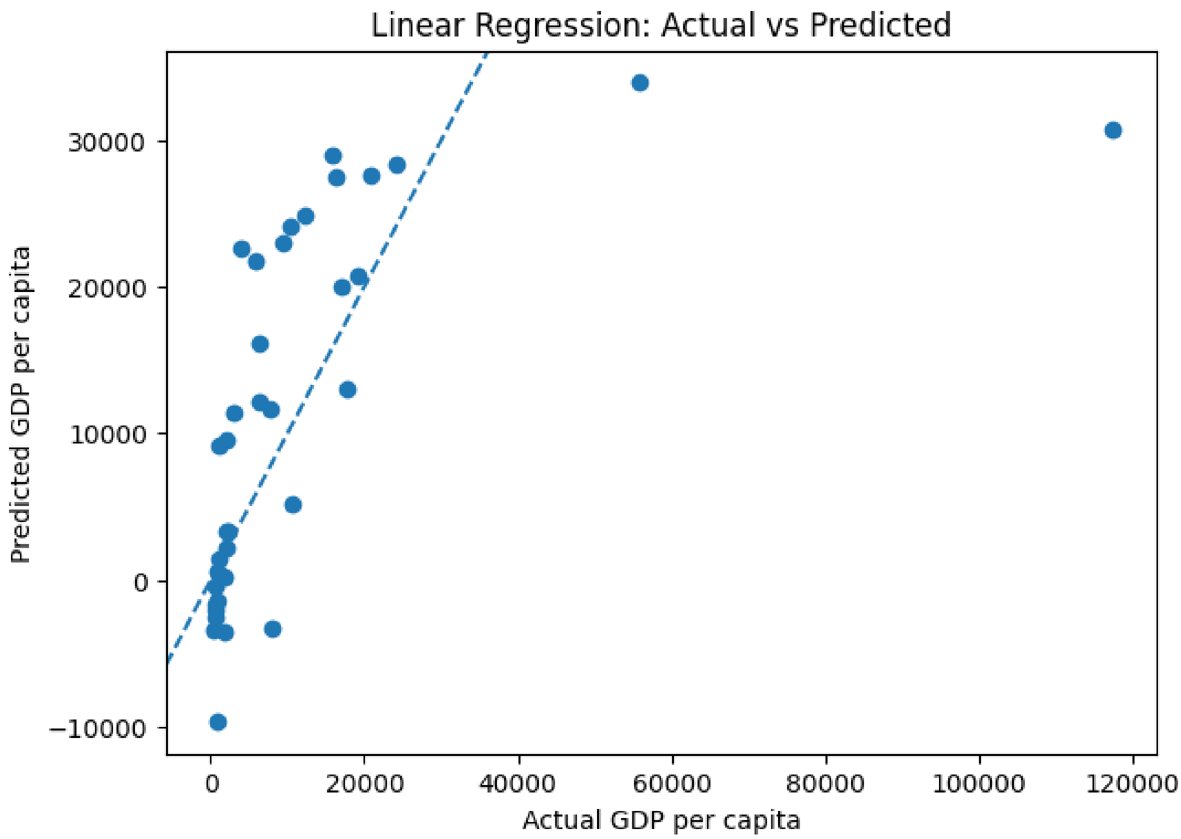
	Feature	Coefficient
0	life_expectancy	8128.025067
5	business_confidence_index	4365.559345
4	log_gdp	3304.097300
3	trade_intensity	1337.734837
2	urban_population_growth	306.898043
1	unemployment	-1964.277272

Appendix C: Figure 13

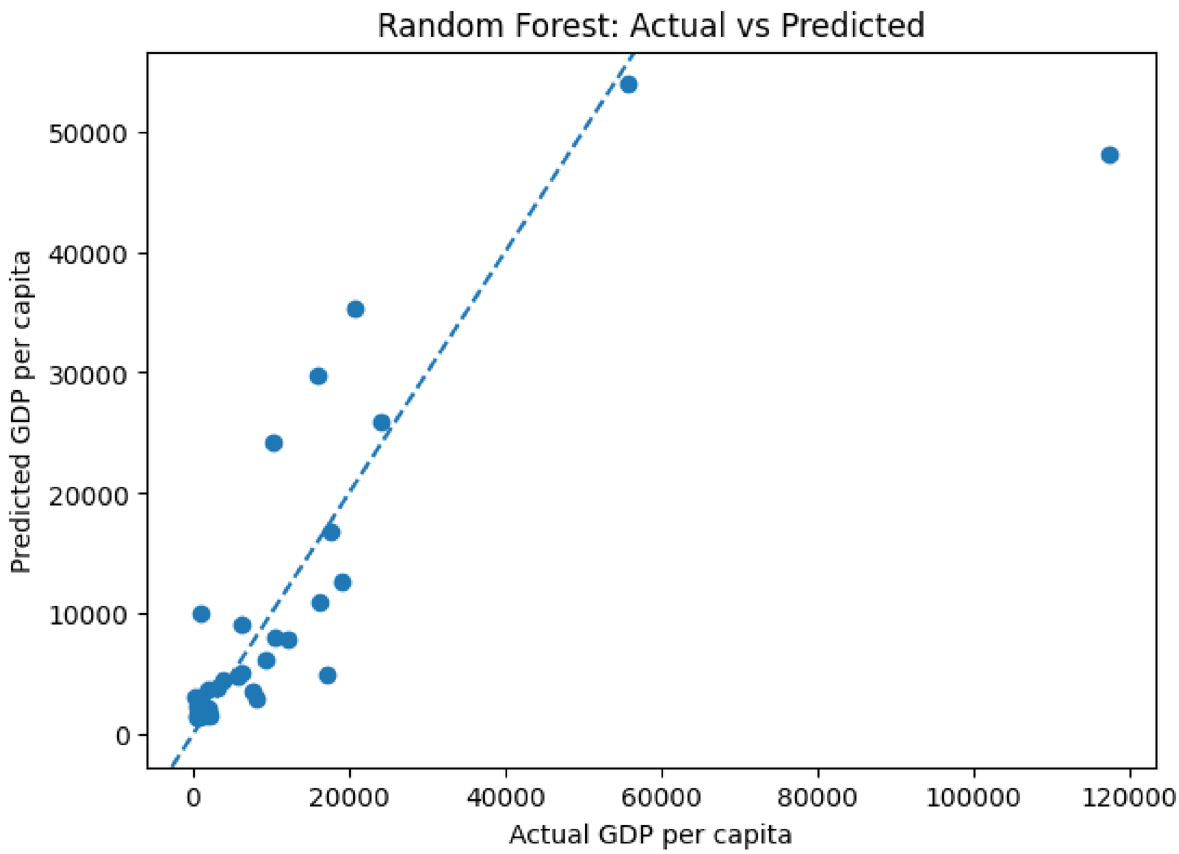
Random Forest Feature Importances

	Feature	Importance
0	life_expectancy	0.815316
4	log_gdp	0.073340
2	urban_population_growth	0.039329
1	unemployment	0.034032
3	trade_intensity	0.028098
5	business_confidence_index	0.009886

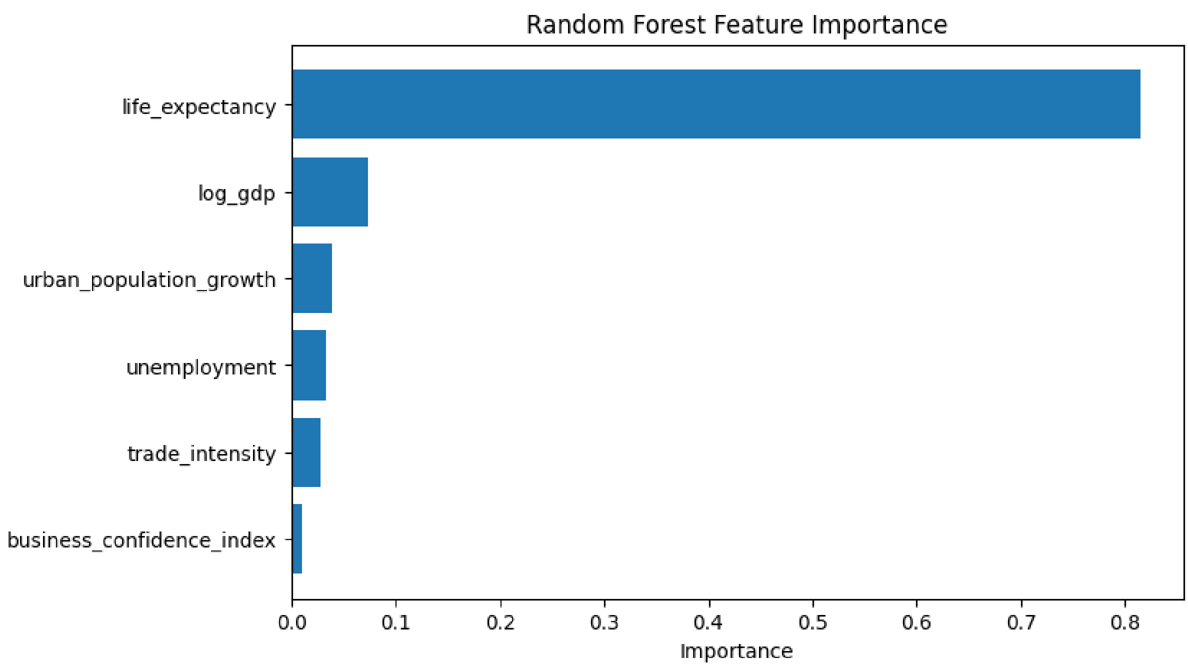
Appendix C: Figure 14



Appendix C: Figure 15



Appendix C: Figure 16



Appendix C: Figure 17

Model C: Unemployment

Target Variable = Unemployment

Model Comparison

	Model	R2	MAE	RMSE
0	Linear Regression	-0.215466	4.502285	5.326559
1	Random Forest	0.096296	3.716867	4.592915

Appendix C: Figure 18

Linear Regression Coefficients

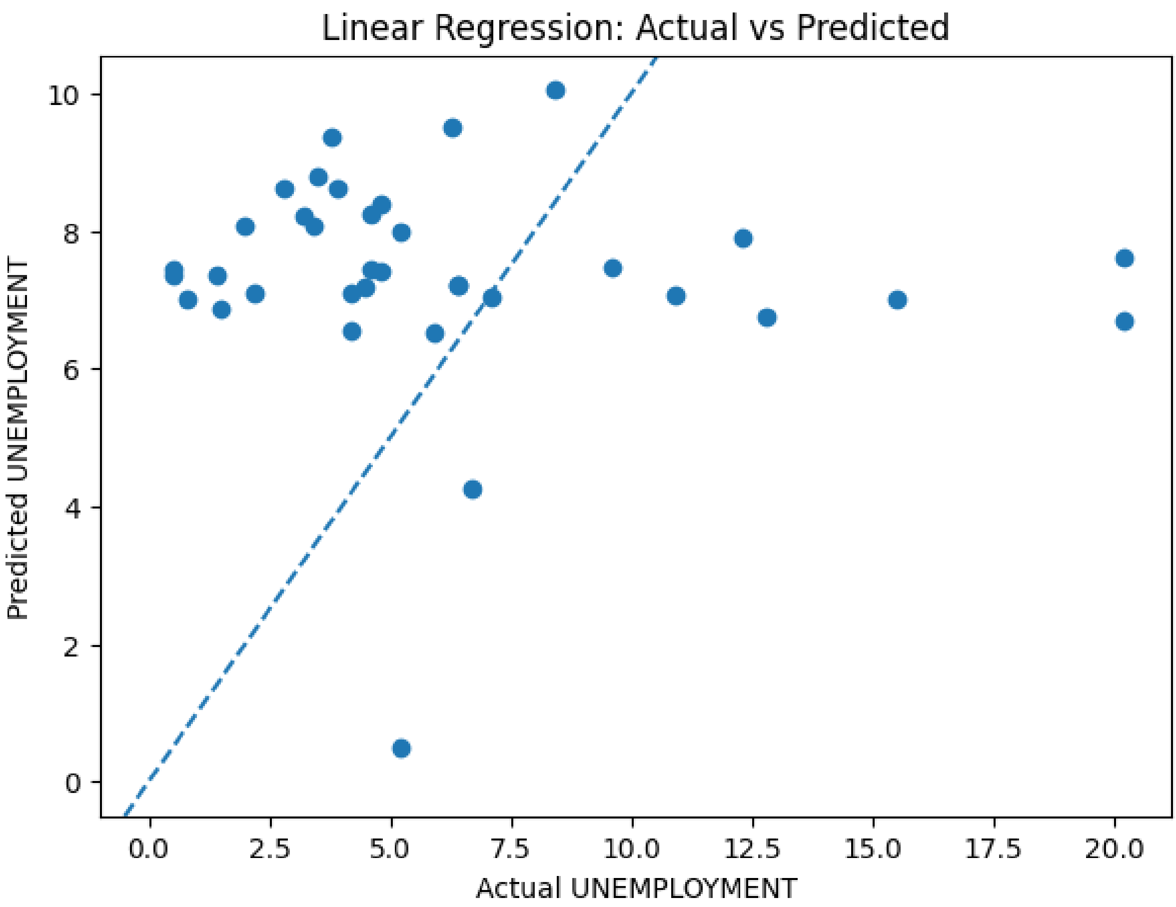
	Feature	Coefficient
5	business_confidence_index	0.618037
3	trade_intensity	0.389209
4	log_gdp	-0.561072
2	life_expectancy	-0.677535
1	urban_population_growth	-0.723793
0	gdp_per_Capita	-1.140136

Appendix C: Figure 19

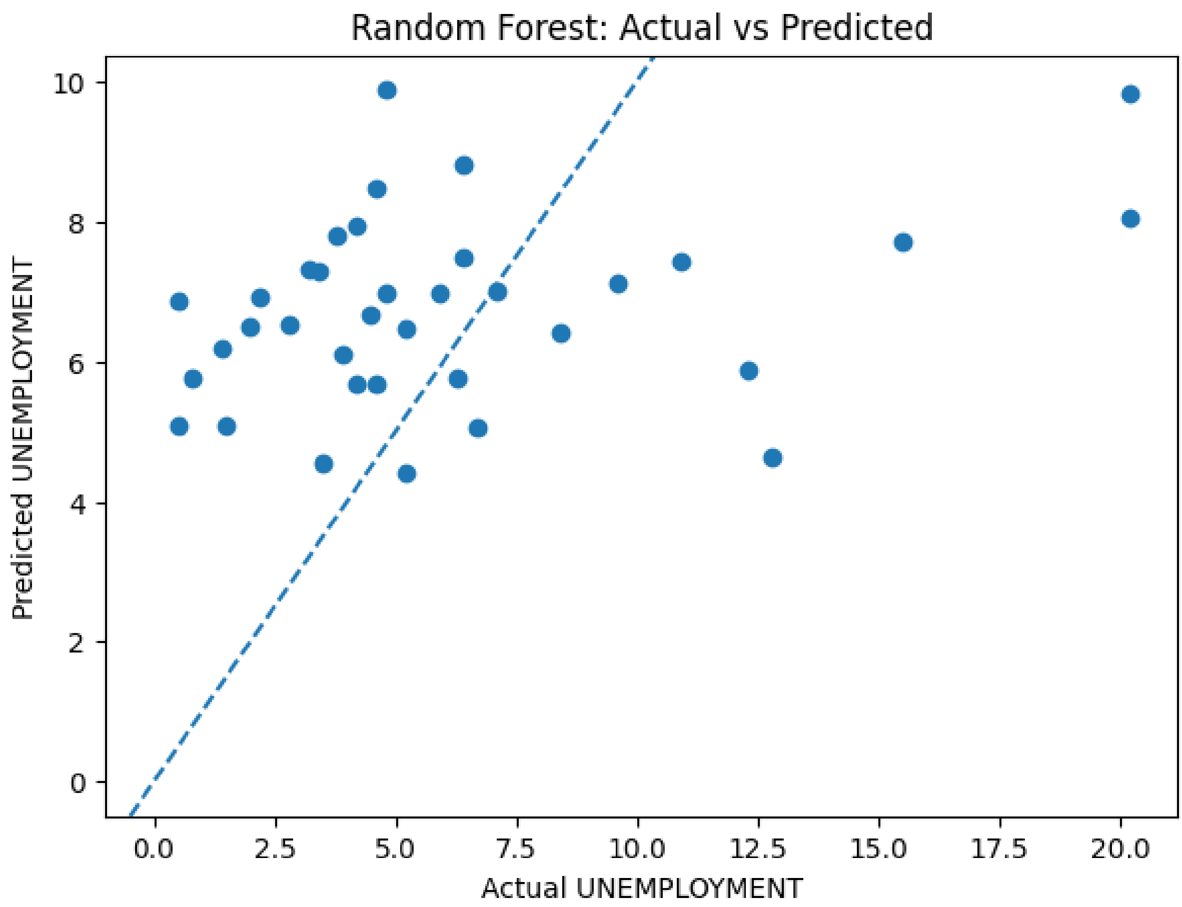
Random Forest Feature Importances

	Feature	Importance
0	gdp_per_Capita	0.267027
2	life_expectancy	0.239903
3	trade_intensity	0.212156
4	log_gdp	0.181745
1	urban_population_growth	0.085995
5	business_confidence_index	0.013175

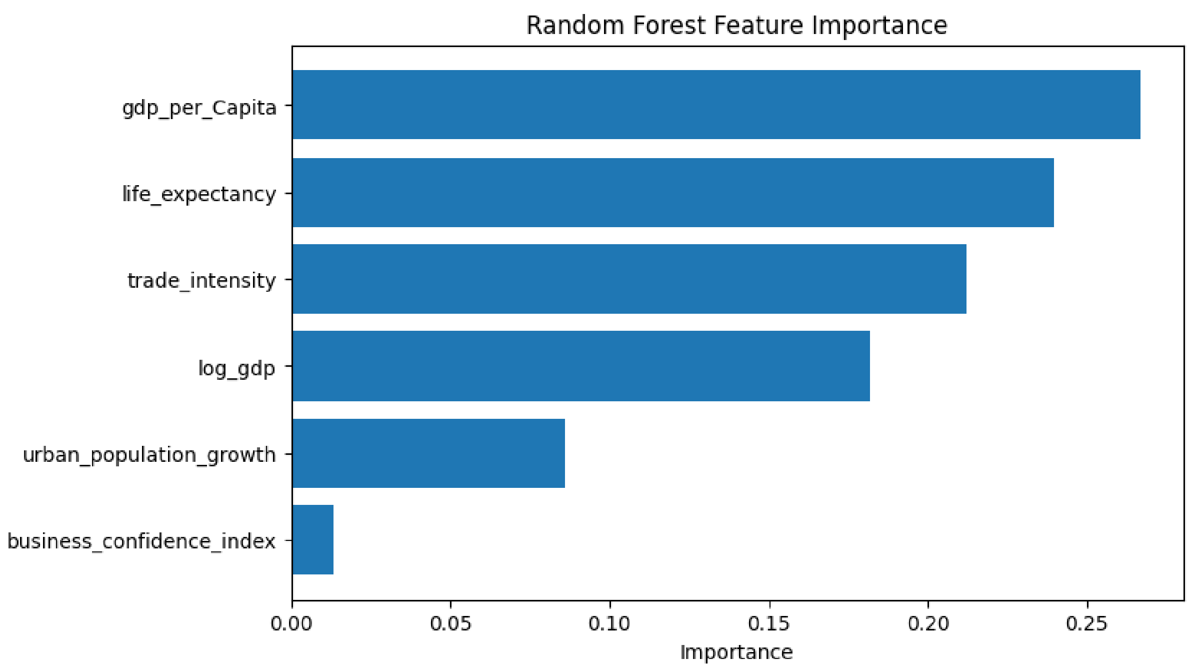
Appendix C: Figure 20



Appendix C: Figure 21



Appendix C: Figure 22



Appendix C: Figure 23